CLOUD COMPUTING

21ISE544

Module – 1: Chapter 1 & 3

- **INTRODUCTION:** Defining Cloud Computing, Cloud Types, Examining the Characteristics of Cloud Computing Historical Developments.
- **VIRTUALIZATION:** Introduction, Characteristics of Virtualized Environments, Taxonomy of Virtualization Techniques, Execution Virtualization, Other Types of Virtualization, Virtualization and Cloud Computing, Pros and Cons of Virtualization, Technology Examples: VMware: Full Virtualization

What is Cloud?

Introduction

- The “cloud” is a term that simply means “the internet.”
- It is a term used to describe a global network of servers, each with a unique function.
- The cloud is not a physical entity,
- It is a vast network of remote servers around the globe which are hooked together and meant to operate as a single ecosystem.

Introduction

- Computing is being transformed into a model consisting of services that are **commoditized**
  - **Allow consumers to make purchasing decisions based solely on the price-tags**
- **It delivered** in a manner similar to utilities such as water, electricity, gas, and telephony
- In such a model, users **access services** based on their **requirements**, regardless of where the services are hosted.

Introduction

- Cloud computing is a **technological advancement** that focuses on the way in which we design
  - **computing systems, develop applications, and leverage existing services for building software**
- Based on concept of ***dynamic provisioning***
  - ***Providing a product or service.***
  - ***Ex:*** ***Services , Compute, Storage , Networking , Information technology (IT)***
- *Pay-per-use* basis from Cloud Computing vendors

Cloud Computing Supporting technologies

- **Web 2.0** – building computing systems
- **Service orientation** – cloud deliver its capabilities
- **Virtualization** – necessary degree of customization, control, and flexibility for building systems

1.1.1 The Vision of Cloud Computing

- Cloud computing allows anyone with a **credit card*** to provision virtual hardware, runtime environments, and services
- These are used for as long as needed, with no up-front commitments required.
- The entire **stack of a computing** system is transformed into a collection of utilities, which can be provisioned and composed together to deploy systems in **hours** rather than days and with virtually **no maintenance costs**.

- This opportunity, initially met with doubt, has now become a practice across several application domains and business sectors
- Vision: IT services are traded as utilities in an open market, without technological and legal barriers
- In this cloud marketplace, cloud service providers and consumers, trading cloud services as utilities, play a central role.

I need to grow my infrastructure, but I do not know for how long...
I have a lot of infrastructure that I want to rent ...
I have a surplus of infrastructure that I want to make use of
I cannot invest in infrastructure, I just started my business.....
I have infrastructure and middleware and I can host applications
Global Cloud Marketplace
I want to focus on application logic and not maintenance and scalability issues
I have infrastructure and provide application services
I want to access and edit my documents and photos from everywhere..

- When plugging an **electric appliance** into an outlet
  - Do we care **how electric power is generated and how it gets to that outlet.? - NO**
- This is because **electricity is virtualized**;
  - i.e., it is readily **available from a wall socket that hides power** generation stations and a huge distribution grid.
- When extended to information technologies, this concept **means delivering useful functions.**
- So it will considered **fully virtualized**, must allow computers to be built from **distributed components** such as **processing, storage, data, and software resources**

- Technologies such as **cluster, grid, and now**, **cloud computing,** have all aimed at allowing access to large amounts of computing power in a fully virtualized manner.
- An important aim of these technologies has been delivering computing **as a utility.**
- **Utility computing** describes a business model for **on-demand delivery** of computing power; consumers pay providers based on usage **("pay-as- you-go").**
- Cloud computing is a category of sophisticated **on-demand computing services.**
  - initially offered by commercial providers, such as Amazon, Google, and Microsoft.

1.1.2 Defining a Cloud:

- The term '**Cloud**' has historically been used in the telecommunications industry as an abstraction of the network in system diagrams.
- Armbrust et al –***"Cloud computing refers to both the applications delivered as services over the Internet, and the hardware and system software in the datacenters that provide those services"***
- **XaaS**: IaaS, SaaS, PaaS

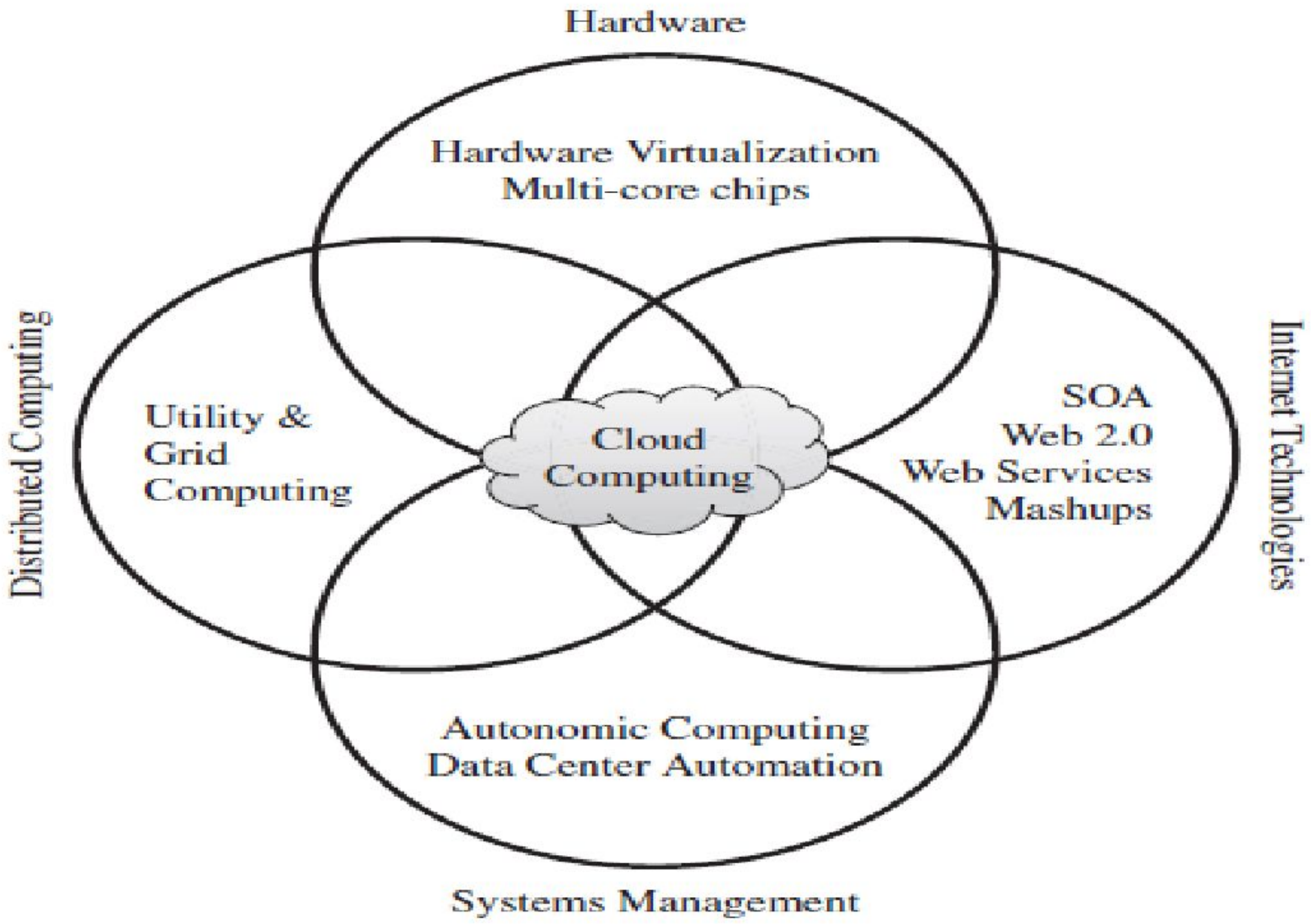
(Infrastructure/Software/Platform as a service.)

- *"Cloud computing is a model for enabling Universal, convenient, on-demand network access to a shared pool of configurable* ***computing resources***
- ***Ex:*** *Networks, Servers, Storage, Applications, and Services - released with minimal management effort or service provider interaction"*
  - ***Pay-per-use***
- It can defined in ***three criteria(I/S/P)***
- The service is accessible via a **Web browser.**
- **Zero capital expenditure** is necessary to get started.
- You pay only for what you use as you use it - ***Pay-per-use*** **.**

1.1.2 Defining a Cloud contd…

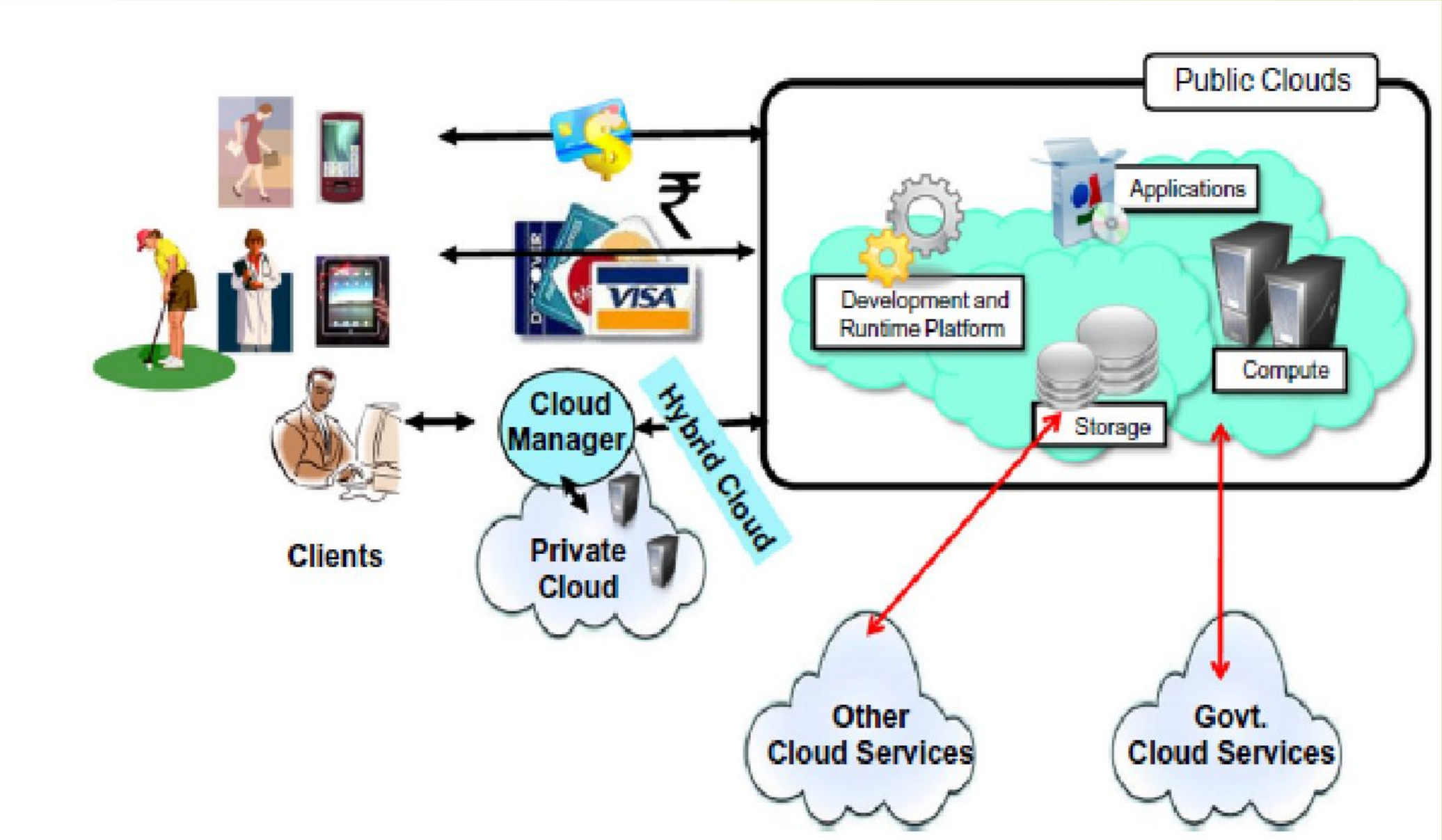
- **Raj kumar Buyya et. al** ----"A cloud is a type of parallel and distributed system consisting of a collection of interconnected and **virtualized** computers that are **dynamically provisioned** and presented as one or more unified computing resources based on "**service-level agreements"** established through "**Negotiation"** between the service provider and consumers".
- SLA = {Negotiated and Agreed QoS parameters +

  rewards + penalties for violation of agreement....}


Hardware
Hardware Virtualization
Multi-core chips
Distributed Computing
Utility &
Grid
Computing
Cloud
Computing
SOA
Web 2.0
Web Services
Mashups
Internet Technologies
Autonomic Computing
Data Center Automation
Systems Management

No capital investments
Cloudbursting
SaaS
Quality of Service
Pay as you go
Internet
PaaS
Green computing
IaaS
Billing
Utility computing
Elasticity
IT outsourcing
Service Level Agreement
Virtual Datacenters
Scalability
IT outsourcing
Privacy & Trust
Provisioning on demand
Security
Virtualization
Cloud Computing?

A Bird's eye view of cloud computing



Deployment models

Public/Internet Clouds

*Third-party, multitenant cloud infrastructure and services

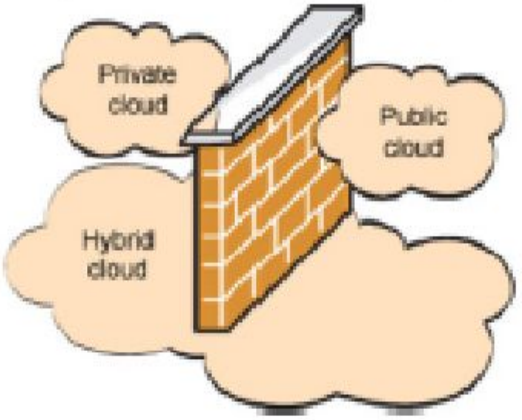
*Available on a subscription basis to all

Private/Enterprise Clouds

*A public cloud model within a company's own datacenter/infrastructure for internal and/or partners' use

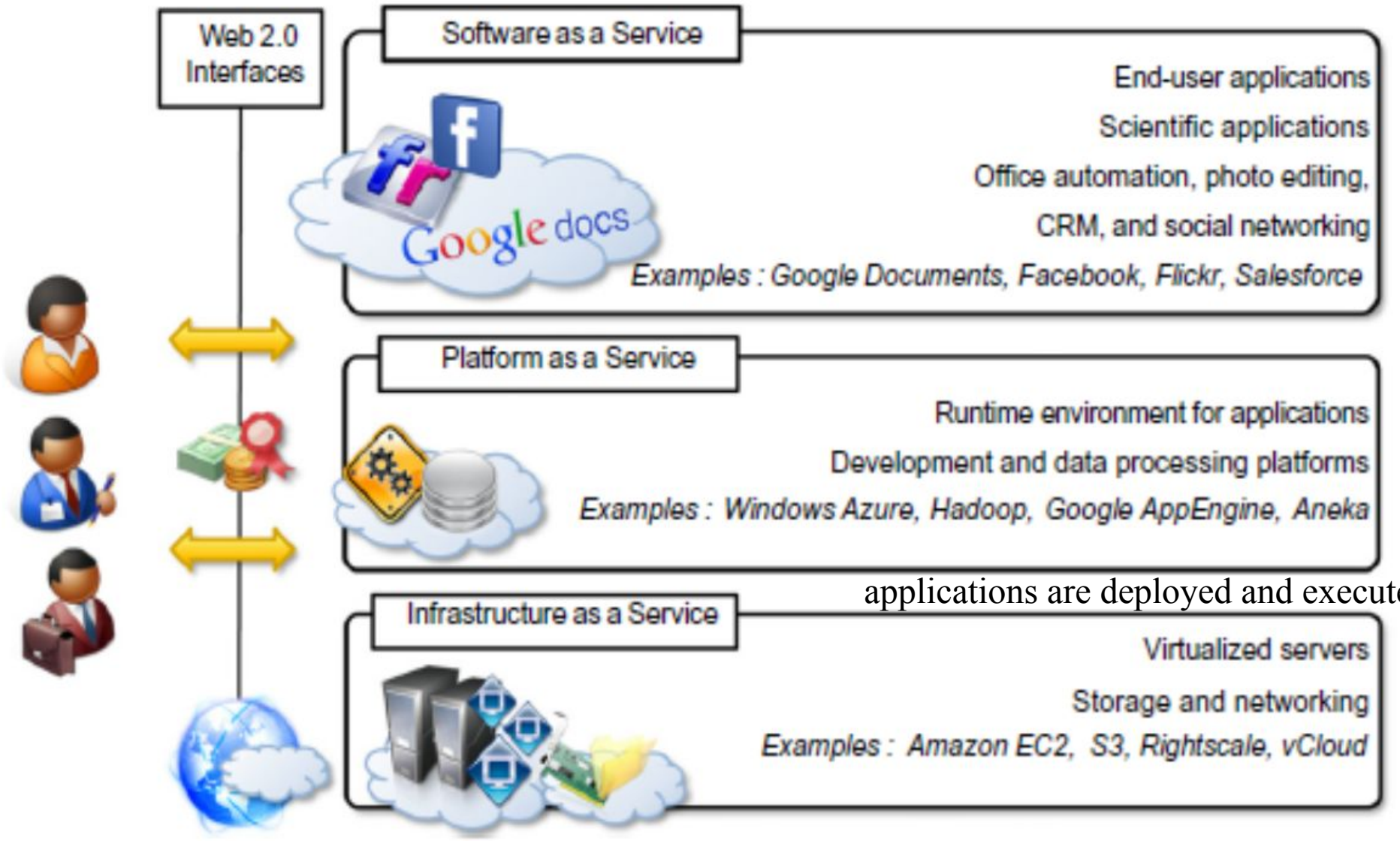
Hybrid/Inter Clouds

* Mixed use of private and public clouds; leasing public cloud services when private cloud capacity is insufficient



1.1.4 Cloud-Computing Reference Model

- A fundamental characteristic of cloud computing is the **capability to deliver a variety of IT services on demand.**
- This variety **creates different views** of what cloud computing is among users
- Even with this lack of uniformity, it is possible to classify cloud computing services offerings into three major categories:
  - *Infrastructure-as-a-Service (**IaaS**)*
  - *Platform-as-a-Service (**PaaS**)* and
  - *Software-as-a-Service (**SaaS**)*



backed by a core middleware platform

applications are deployed and executed.

VIRTUAL MACHINE INSTANCES

Layered view- computing stack

FIGURE 1.5

The Cloud Computing Reference Model.

- The model organizes the wide range of cloud computing services into a **layered view** that walks the computing stack from **bottom to top**.
- At the base of the stack, ***Infrastructure-as-a-Service*** solutions deliver **infrastructure on demand** in the form of **virtual *hardware*, *storage*, and *networking*.**
- Virtual hardware is utilized to provide compute on demand in the form of "***virtual machine instances***".
- These are created at **users request** on the provider's infrastructure, and users are given **tools and interfaces** to **configure the software stack installed in the virtual machine**.
- The **pricing** model is usually defined in terms of dollars per hour.

- ***Platform-as-a-Service*** solutions are the next step in the stack.
- They deliver **scalable and elastic runtime environments on demand** and host the execution of applications.
- These services are backed by a core **middleware platform** that is responsible **for creating the abstract environment** where applications are deployed and executed.
- It is the responsibility of the service provider to provide **scalability and to manage fault tolerance**, while ***users are requested to focus on the logic of the application developed by leveraging*** the provider's APIs and libraries.

- At the top of the stack, ***Software-as-a-Service*** solutions provide applications and services on demand.
- Most of the common functionalities of desktop applications—such as **office automation**, **document management**, **photo editing, and customer relationship management (CRM).**
- These software's are **replicated** on the provider's infrastructure and made more **scalable and accessible through a browser on demand.**
- These applications are **shared across multiple users** whose interaction is isolated from the other users.
- The SaaS layer is also the area of **social networking Websites**, which influence cloud-based infrastructures to sustain the load generated by their popularity

1.1.5 Characteristics and Benefits

Cloud computing has some interesting characteristics that bring benefits to both Cloud Service Consumers (CSCs) and Cloud Service Providers (CSPs).

These characteristics are

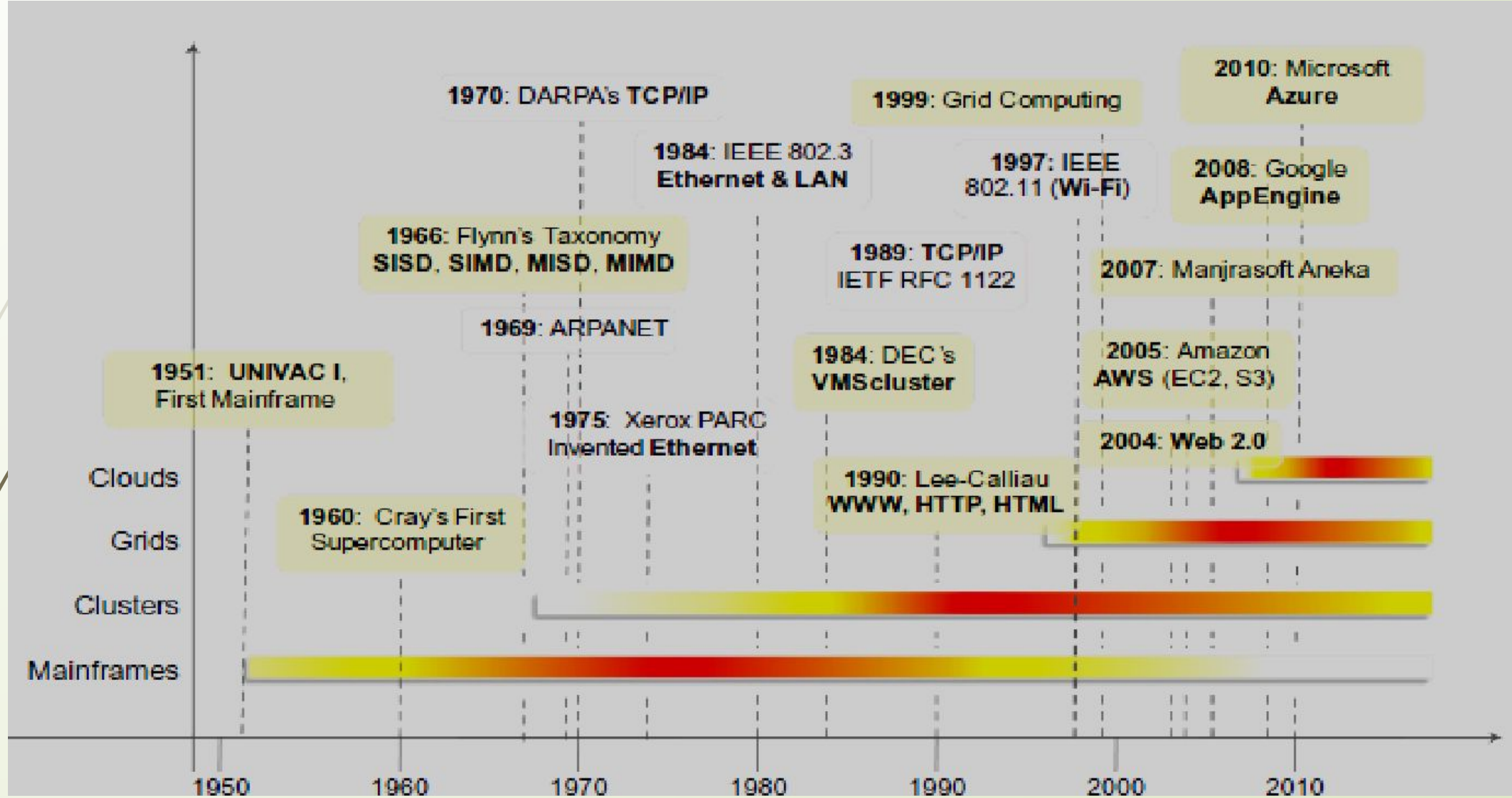
- No up-front commitments
- On-demand access
- Nice pricing
- Simplified application acceleration and scalability
- Efficient resource allocation
- Energy efficiency
- Seamless creation and use of third-party services

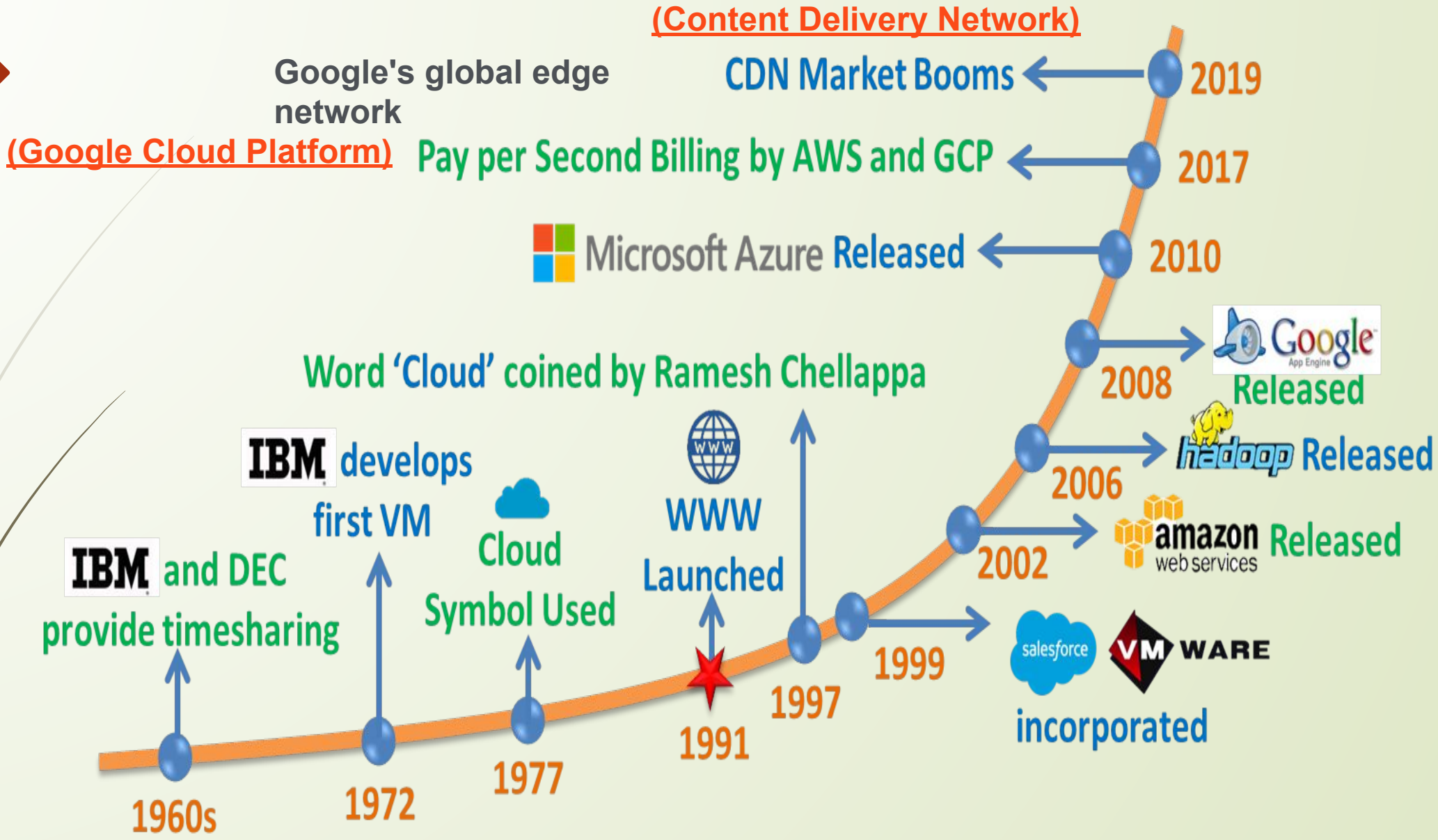
1.1.6 Challenges ahead

- Challenges concerning the ***dynamic provisioning*** of cloud computing **services** and **resources**.
- Technical challenges also arise for cloud service providers for the management of large computing infrastructures and the use of virtualization technologies on top of them
- **Security**
  - Confidentiality, secrecy
  - Protection of data
  - Legal issues
    - Data centers in different geographical locations.

- Security in terms of **confidentiality, secrecy, and protection of data** in a cloud environment is another important challenge.
- Organizations do not own the infrastructure they use to **process data and store information**.
- This condition poses challenges for **confidential data**, which organizations cannot afford to reveal.
- Legal issues may also arise.
- These are specifically tied to the universal nature of cloud computing, which spreads computing **infrastructure across diverse** geographical locations.

1.2 Historical Developments




(Content Delivery Network)
Google's global edge network
CDN Market Booms
2019
(Google Cloud Platform)
Pay per Second Billing by AWS and GCP
2017
Microsoft Azure Released
2010
Google App Engine
2008
Released
Word 'Cloud' coined by Ramesh Chellappa
IBM develops first VM
WWW
hadoop Released
2006
Cloud Symbol Used
Launched
amazon web services Released
2002
IBM and DEC provide timesharing
salesforce
VMWARE
1999
1997
incorporated
1991
1977
1972
1960s

Historical developments

- The idea of renting computing services by maximizing the **distributed computing facilities**.
- This process has created **a series of favorable conditions** for the realization of cloud computing
- provides an over view of the evolution of the distributed computing technologies that have influenced cloud computing.

- There are **Five core technologies** that played an important role in the realization of cloud computing.

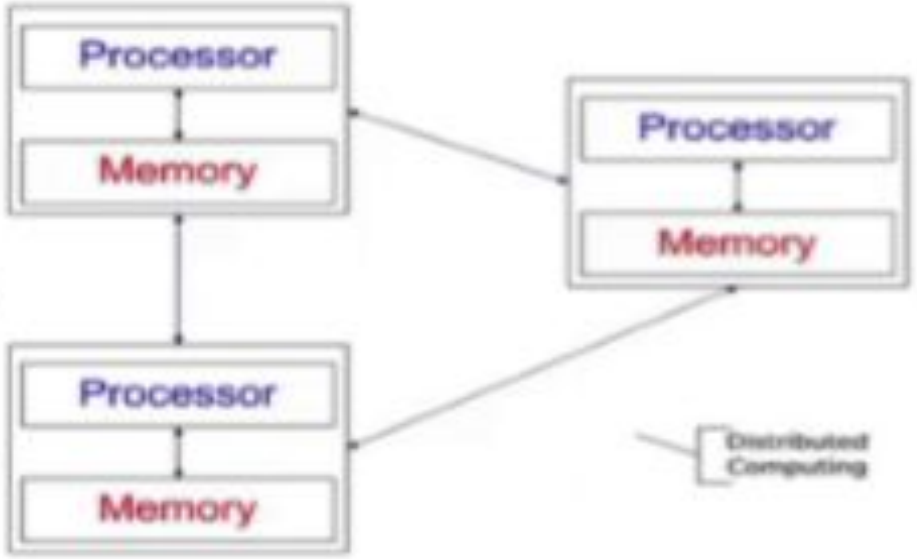
1. Distributed systems,
2. Virtualization,
3. Web2.0
4. Service orientation, and
5. Utility computing.

1.2.1 Distributed systems

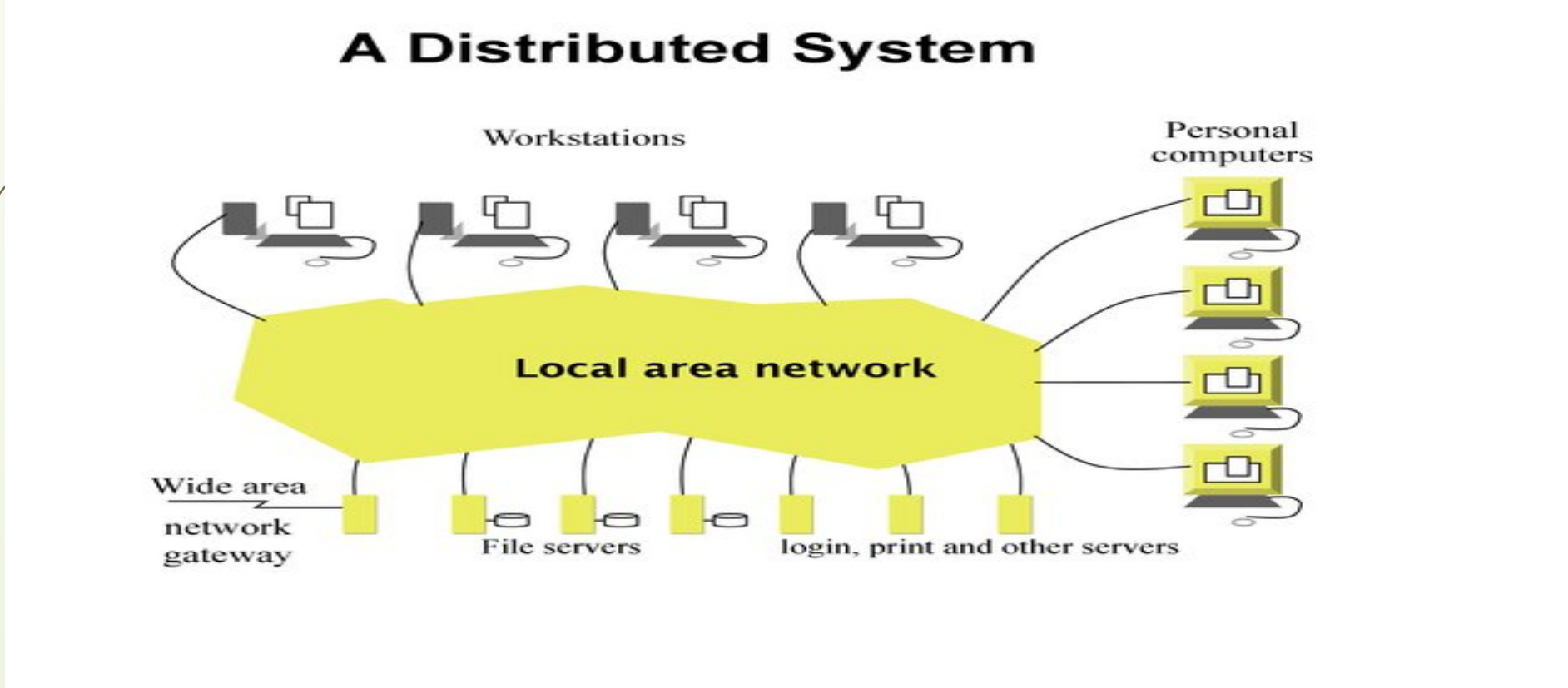
- Distributed cloud computing splits one task across multiple computers at different locations—all of which are networked.
- Each computer will complete an aspect of the task, allowing for the task to be completed faster.

Distributed Computing/System?

- Distributed computing
  - Field of computing science that studies distributed system.
  - Use of distributed systems to solve computational problems.
- Distributed system
  - Wikipedia
    - There are several autonomous computational entities, each of which has its own local memory.
    - The entities communicate with each other by message passing.
  - Operating System Concept
    - The processors communicate with one another through various communication lines, such as high-speed buses or telephone lines.
    - Each processor has its own local memory.



- A **distributed system** is a collection of independent computers that appears to its users as a single coherent system.



Example Distributed Systems

- Internet
- ATM (bank) machines
- Intranets/Workgroups
- Computing landscape will soon consist of ubiquitous network-connected devices

Computers in a Distributed System

- *Workstations*: Computers used by end-users to perform computing
- *Server Systems*: Computers which provide resources and services
- *Personal Assistance Devices*: Handheld computers connected to the system via a wireless communication link.

Common properties of Distributed Computing

- Fault tolerance
  - When one or some nodes fails, the whole system can still work fine except performance.
  - Need to check the status of each node
- Each node play partial role
  - Each computer has only a limited, incomplete view of the system.
  - Each computer may know only one part of the input.
- Resource sharing
  - Each user can share the computing power and storage resource in the system with other users
- Load Sharing
  - Dispatching several tasks to each nodes can help share loading to the whole system.
- Easy to expand
  - We expect to use few time when adding nodes. Hope to spend no time if possible.
- Performance
  - Parallel computing can be considered a subset of distributed computing

Why Distributed Computing?

- Nature of application
- Performance
  - Computing intensive
    - The task could consume a lot of time on computing. For example, Computation of Pi value using Monte Carlo simulation
  - Data intensive
    - The task that deals with a large amount or large size of files. For example, Facebook, LHC(Large Hadron Collider) experimental data processing.
- Robustness
  - No SPOF (Single Point Of Failure)
  - Other nodes can execute the same task executed on failed node.

- The primary purpose of distributed systems is **to share resources and utilize them better.**
- Distributed systems often exhibit other properties such as
    1. Heterogeneity,
    2. Openness,
    3. Scalability,
    4. Transparency,
    5. Concurrency,
    6. Continuous Availability, and
    7. Independent failures.

- Three major milestones have led to cloud computing:
  - Mainframes.
  - Clusters.
  - Grids.

- Mainframes
  - Large computational facilities, Massive IO operations
  - Large organizations - - Online banking, airline ticket booking, supermarket, etc.,

- **Mainframes/Super computers:** These were the first examples of large computational facilities taking the advantages of processing units.
- Mainframes were **powerful**, **highly reliable** computers specialized for large data movement and massive input/output (I/O) operations.
- They were mostly used by large organizations for **bulk data processing tasks.**
- such as **online transactions**, **enterprise resource planning**, and other operations involving the processing of significant amounts of data.

- One of the most attractive features of mainframes was the **ability to be highly reliable computers that were "always on" and capable of tolerating failures transparently**.
- **No system shutdown** was required to **replace failed** components, and the system could work without interruption.
- **Now their popularity and deployments have reduced**, but evolved versions of such systems are still in use for transaction processing
- such as online banking, airline ticket booking, supermarket and telecommunications, and government services.

- **Clusters.** Cluster computing started as a **low-cost alternative** to the use of **mainframes and supercomputers**.
- These machines could then be connected by a **high-bandwidth network** and **controlled by specific software tools** that manage them as a single system.
- Built by **commodity machines**, they were cheaper than mainframes and made **high-performance** computing available to a large number of groups.
    - Like universities and small research labs.
- Moreover, clusters could be easily extended if more computational power was required

- **Grids.** Grid computing appeared in the early 1990s as an evolution of cluster computing.
- Users can **“consume”** resources in the same way as they use other utilities such as power, gas, and water.
- Grids initially developed **as aggregations of geographically dispersed clusters by means of Internet connections.**
- These clusters belonged to different organizations, and arrangements were made among them to share the computational power.

1.2.2 Virtualization

- Virtualization is a process that allows a computer to **share** its **hardware resources** with multiple digitally separated environments.
- In other words, Virtualization is a technique, which allows to **share a single physical instance** of a resource or an application among **multiple customers and organizations.**
- Each virtualized environment runs within its allocated resources, such as memory, processing power, and storage.

1.2.2 Virtualization

- It is a **Fundamental technology** that powers cloud computing
- Allowing the **concept** of **emulation of IT infrastructure like** hardware, runtime environments, storage, and networking.
- ***Hardware virtualization*** - allows the coexistence of different **software stacks** on top of the same hardware.
  - E.g., Amazon EC2, RightScale, VMware vCloud
- ***Storage and network virtualization:*** ***Amazon s3***
- ***Process virtual machines:*** Google AppEngine and Windows Azure

1.2.3 Web 2.0

- The Web is the **primary interface** through which cloud computing **delivers its services**
- Web 2.0 - a rich platform for application development
- Web 2.0 brings **interactivity** and **flexibility** into Web pages, providing enhanced user experience by gaining Web-based access to all the functions
- These capabilities are obtained by **integrating a collection of standards and technologies.**
- Examples of Web 2.0 applications are Google Documents, Google Maps, Flickr, Facebook, Twitter, YouTube, Blogger, and Wikipedia

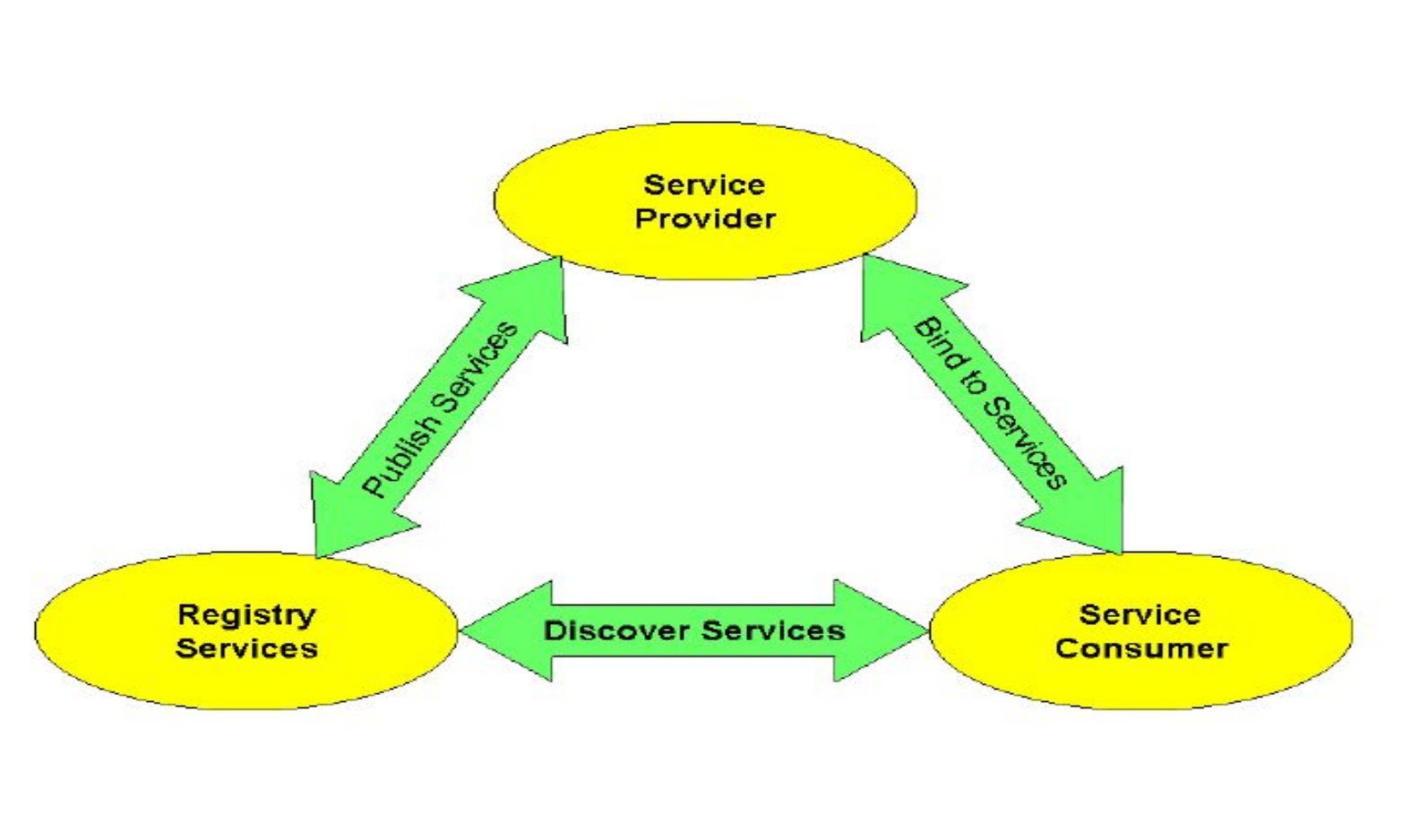
1.2.4 Service-oriented Computing

- It is the model for distributed computing to **enable application networks** to be built **using services as fundamental elements** for developing applications.
- A good example of a SOA-based system is a set of customer services, like CRM, ERP, Product Information Management System (PIM), etc.
- These services can be implemented using different technologies and support diverse protocols of communication, data models, etc.

1.2.4 Service-oriented Computing

- ***Service-oriented computing (SOC)*** supports the development of rapid, low-cost, flexible, interoperable, and evolvable ***applications*** and ***systems***
- A **service** is an abstraction representing a self-describing and platform diagnostic **component** that can perform any function.
- Means any thing from a **simple function** to a **complex business** process
- Characteristics-
  - Loosely coupled
  - reusable
  - programming language independent, and
  - location transparent

- Services are composed and aggregated into a service-oriented architecture (SOA)



- SOC introduces two important concepts ---
- **QoS** requirements are established between the client and the provider via an SLA that identifies the minimum values (or an acceptable range) for the QoS attributes that need to be satisfied upon the service call
- Response time, availability , scalability etc.,,
- **SaaS**

1.2.5 Utility Computing

- Utility computing is a vision of computing that **defines service- provisioning model** for compute services in which resources such as **storage, compute power, applications**, and **infrastructure** are packaged and **offered** on a **pay-per-use** basis.
- The idea of providing computing as a utility like natural gas, water, power, and telephone.

1.4 Computing platforms and technologies

1. *Amazon web services (AWS)*
2. *Google AppEngine*
3. *Microsoft Azure*
4. *Hadoop*
5. *Force.com and Salesforce.com*
6. *Manjrasoft Aneka*

- **AWS** offers comprehensive cloud IaaS services ranging from **virtual compute**, **storage**, and **networking** to complete computing stacks.
- AWS is mostly known for its **compute and storage** on demand services, namely **Elastic Compute Cloud (EC2)** and **Simple Storage Service (S3)**.
- EC2 provides users with customizable virtual hardware that can be used as the base infrastructure for deploying computing systems on the cloud.
- It is possible to choose from a large variety of virtual hardware configurations, including GPU and cluster instances.
- EC2 instances are deployed either by using the AWS console, which is a comprehensive Web portal for accessing AWS services, or by using the Web services API available for several programming languages.
- Besides EC2 and S3, a wide range of services can be leveraged to build **virtual computing systems. including networking support, caching systems, DNS, database (relational and not) support, and others.**

- Google AppEngine is a **scalable runtime environment** devoted to **executing Web applications**.
- These take advantage of the large computing infrastructure of Google.
- AppEngine provides **both a secure execution environment and a collection of services** that simplify the **development of scalable and high-performance Web applications.**
- Developers can build and test applications on their own machines using the **AppEngine software development kit (SDK**.
- Once development is complete, **developers can easily migrate their application to AppEngine,** set quotas to contain the costs generated, and make the application available to the world.
- The languages **currently supported are Python, Java, and Go.**

- **Microsoft Azure** is a cloud **operating system and a platform for developing** applications in the cloud.
- It provides a **scalable runtime environment for Web applications and distributed applications in general.**
- Applications in Azure are organized around the concept of roles currently, there are three types of role: **Web role, worker role, and virtual machine role.**
- The Web role is designed to **host a Web application,**
- The worker role is a **more generic container of applications** and can be used to **perform workload processing**, and
- The **virtual machine role provides a virtual environment**
- Besides roles, Azure provides a set of additional services that complement application execution, such as **support for storage (relational data and blobs), networking, caching, content delivery, and others**

- Apache Hadoop is an **open-source framework that is suited for processing large data** sets on commodity hardware.
- Hadoop is an implementation of **MapReduce, an application programming model developed by Google**, which provides two fundamental operations for data processing: map and reduce.
- The former **transforms and synthesizes the input data provided by the user; the latter aggregates the output obtained by the map operations.**
- Hadoop provides the **runtime environment, and developers need only provide the input data and specify the map and reduce functions that need to be executed.**
- **Yahoo!, the sponsor of the Apache Hadoop project**, currently, Yahoo! manages the largest Hadoop cluster in the world, which is also available to academic institutions
- **Hadoop is an integral part of the Yahoo! cloud infrastructure and supports several business processes of the company**.

- **Force.com and Salesforce.com**
- Force.com is a cloud computing platform for developing **social enterprise applications. It is a Software-as-a-Service solution for customer relationship management.**
- Force.com allows developers to create applications by composing ready-to-use blocks; a complete set of components supporting all the activities of an enterprise are available.
- The platform provides complete support for developing applications, from the design of the data layout to the definition of business rules and workflows and the definition of the user interface.
- The Force.com platform is **completely hosted on the cloud and provides complete access to its functionalities and those implemented in the hosted applications through Web services technologies.**

- **Manjrasoft Aneka**
- Manjrasoft Aneka is a **cloud application platform for rapid creation of scalable applications and their deployment** on various types of clouds in a seamless and elastic manner.
- It supports for **developing applications and a distributed runtime environment** that can be deployed on **heterogeneous hardware (clusters, networked desktop computers, and cloud resources).**
- Developers can choose different abstractions to design their application: **tasks, distributed threads, and map-reduce**.
- These applications are then executed on the distributed service-oriented runtime environment, which can **dynamically integrate additional resource on demand.**

Virtualization

- Virtualization is a process that allows a computer to share its hardware resources with **multiple digitally separated environments**.
- In other words, Virtualization is a **technique**, which allows **to share a single physical instance of a resource or an application** among **multiple customers and organizations.**
  Ex: Google Photos, storage, etc.,
- It does by assigning a logical name to a physical storage and providing a pointer to that physical resource when demanded.

VIRTUALIZATION

- Virtualization technology is one of the **fundamental components** of cloud computing, especially in regard to **infrastructure based services(IaaS)**.
- Virtualization allows the creation of a **secure**, **customizable**, and **isolated execution environment** for **running applications**, without affecting other users' applications.
- Virtualization is a large **umbrella** of **technologies** and **concepts** that are meant to provide an conceptual environment.
- It may be a **virtual hardware** or an **operating system** to run applications.

- The ***hardware virtualization***, which plays a fundamental role in efficiently delivering ***Infrastructure-as-a-Service*** **(IaaS)** solutions for cloud computing.
- virtualization technologies available in many **flavors.**
  1. **The operating system level**,
  2. **The programming language level**, and
  3. **The application level**.
- Moreover, virtualization technologies provide a virtual environment for **not only executing applications** but also for **storage, memory, and networking**.

- Virtualization is widely used to deliver **customizable computing environments on demand to** provides a **elastically scalable systems** that can provision **additional capability** with **minimum costs**.
- Virtualization technologies have gained importance/ popularity due to the union of several phenomena:

1. **Increased performance and computing capacity .**
2. **Underutilized hardware and software resources**
3. **Lack of space.**
4. **Greening initiatives**
5. **Rise of administrative costs**

1. Increased performance and computing capacity

- Nowadays, the average end-user **desktop PC** is **powerful enough** to **meet** almost all the **needs** of everyday **computing**, with **extra capacity** that is **rarely used**.
- Almost all these **PCs have resources** enough to **host** a **virtual machine** manager and **execute a virtual machine** with by far **acceptable performance**.
- Where **supercomputers** can provide immense compute power that can accommodate the execution of **hundreds or thousands** of virtual machines.

2. Underutilized hardware and software resources

- Hardware and software underutilization is occurring due to following reasons.

  (1) Increased performance and computing capacity, and

  (2) The effect of limited or periodic use of resources.

Computers today are so **powerful** that in most cases only a **fraction of their capacity is used** by an **application** or the **system**.

Moreover, if we consider the **IT infrastructure of an enterprise**, many computers are only **partially utilized.**

Whereas they could be used **without interruption** on a **24/7/365** basis.

For example:

Desktop PCs mostly devoted to office automation tasks and used by administrative staff are only used during work hours, remaining completely unused overnight.

Using these resources for other purposes **after hours could improve the efficiency of the IT infrastructure**.

To transparently provide such a service, it would be necessary to deploy a **completely separate environment**, which can be **achieved through virtualization**.

3. Lack of space.

- The continuous need for **additional capacity**, whether **storage or compute power**, makes **data centers** grow quickly.
  - Ex: **Google and Microsoft** have their infrastructures by building **data centers as large as football fields** that are able to **host thousands of nodes**.
- Although this is **feasible** for IT companies, in most cases **enterprises cannot afford to build** another data center to accommodate additional resource capacity.
- This condition, along with **hardware underutilization**, has led to the distribution of a technique called ***server consolidation***, for which virtualization technologies are fundamental.

Information:

- Google has nearly **30 owned data centers** either in operation or under development across 10 countries throughout the world.
- Azure global infrastructure : The physical component is comprised of **200+ physical datacenters**, arranged into regions, and linked by one of the largest interconnected networks on the planet.

Microsoft Data Centers

Google Data Centers

4. Greening initiatives.(Go-GREEN)

- Recently, companies are increasingly looking for ways to **reduce the amount of energy** they **consume** and to **reduce their carbon footprint**.
- Data centers are one of the major power consumers;
  - Maintaining a data center operation not only involves keeping servers on,
  - But a great deal of energy is also consumed in keeping them cool.

- Infrastructures for **cooling i.e., AC** have a **significant impact** on the **carbon footprint of a data center**.
- Hence, **reducing** the number of **servers** through **server consolidation** will **definitely reduce** the impact of cooling and **power consumption of a data center**.
- Virtualization technologies can provide an **efficient way of consolidating servers.**

5. Rise of administrative costs

- *The* increased **demand for additional capacity**, which translates into more servers in a data center, is also **responsible for a significant increment in administrative costs**.
- Computers in particular, servers do not operate all on their own, but they require **care and feeding from system administrators.**
- **Common system administration** tasks include
  - Hardware monitoring,
  - Defective hardware replacement,
  - Server setup and updates,
  - Server resources monitoring, and backups.

- These are **labor-intensive** operations
- The higher the number of servers that have to be managed, the higher the **administrative costs**.
- ***Virtualization can help reduce the number of required servers for a given workload***
- Thus reducing the cost of the administrative personnel

Characteristics of virtualized environments

- Virtualization is a broad concept that refers to the *creation of a virtual version of something*, whether **hardware**, a **software** environment, **storage**, or a **network**.
- In a virtualized environment there are three major components:
  1. ***Guest*,**
  2. ***Host*, and**
  3. ***Virtualization layer*.**

- **Host**: It represents the original environment
- **Guest**: It is supposed to be managed.
- ***virtualization layer:*** ***It*** is responsible for recreating the same or a different environment where the guest will operate

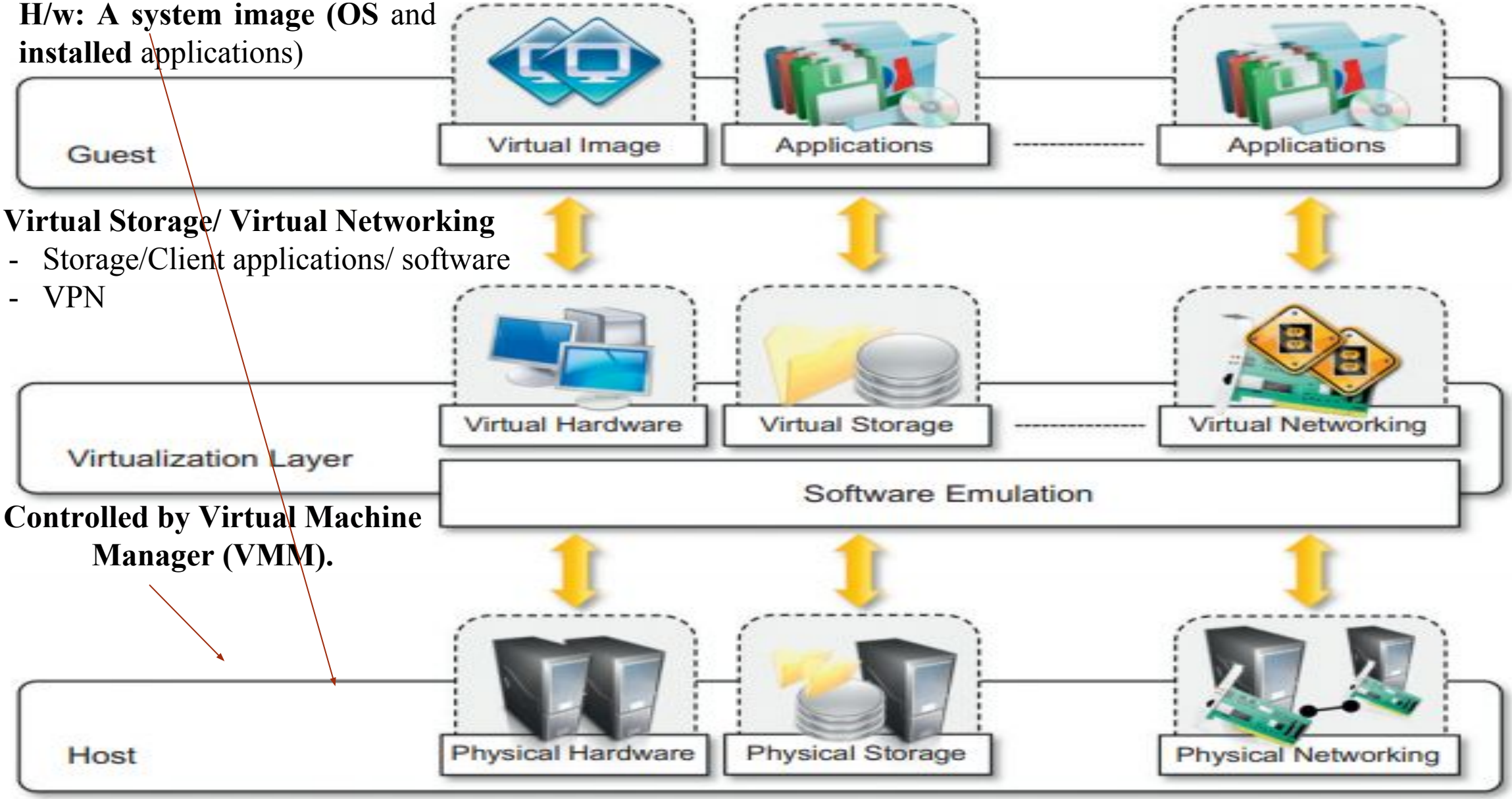
The virtualization reference model

H/w: A system image (OS and **installed** applications)

Virtual Storage/ Virtual Networking

- Storage/Client applications/ software
- VPN

Controlled by Virtual Machine Manager (VMM).



- In the case of **hardware virtualization**, the **guest** is represented **by a system image (**An **OS** and **installed** applications)
- These are installed on top of **virtual hardware** that is **controlled and managed** by the **virtualization layer**, also called the **virtual machine manager (VMM).**
- The host is instead represented by the **physical hardware/ the operating system,** that defines the **environment** where the virtual machine manager is **running**.
- In the case of **virtual storage**, the guest might be **client applications** or users that interact with the virtual storage management software deployed on top of the real storage system.

- The case of virtual networking is also similar:
- The guest applications and users interacts with a virtual network,
- Such as a virtu**al private network (VPN),** which is managed by specific software (VPN client) using the physical network available on the node.
- VPNs are useful for creating the illusion of being within a different physical network and thus accessing the resources in it, which would otherwise not be available.
- The main common characteristic of all these different implementations is the fact that the virtual environment is created by means of a **software program.**
- The ability to use **software to emulate** such a wide variety of environments creates a lot of opportunities

Advantages of Virtualization solutions

- **Increased security**
- **Managed execution**
- **Portability**

Increased security

- The ability to control the execution of a guest in a completely **transparent manner** opens new possibilities for delivering a **secure, controlled execution environment**.
- The virtual machine represents an **emulated environment** in which the guest is executed.
- All the operations of the guest are generally performed against the **virtual machine**, which **then translates and applies them to the host.**
- Sensitive information that is contained in the host can be **naturally hidden**.

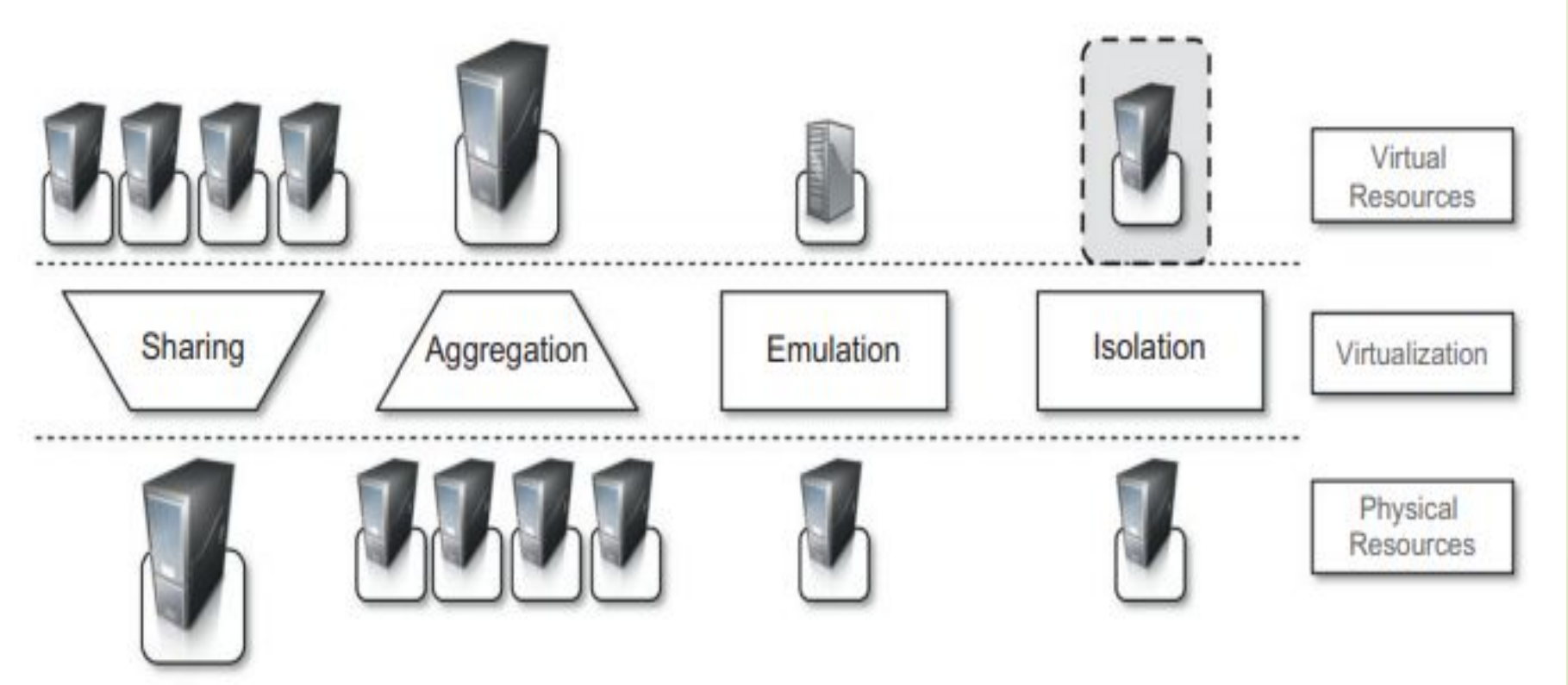
Managed execution

- Virtualization of the execution environment not only allows increased security, but a wider range of features also can be implemented.
- In particular, ***sharing***, ***aggregation***, ***emulation***, **and** ***isolation*** are the most relevant features.

Aggregation:

- A group of separate hosts can be **tied together** and represented to guests as a single virtual host.
- This function is naturally implemented in **middleware for distributed computing,** with a classical example represented by cluster management software

Functions enabled by managed execution



Emulation:

- In **computing**, an **emulator** is hardware or software that enables one computer system (called the host) **to behave like another computer system** (called the guest).
- **Emulation** refers to the ability of a computer program in an electronic device to **emulate** (or imitate) another program or device.
- This allows for **controlling and tuning** the environment that is exposed to guests.
- Hardware virtualization solutions are able to provide virtual hardware and emulate a particular kind of device such as *Small Computer System Interface (SCSI)* devices for file I/O, without the hosting machine having such hardware installed

Isolation:

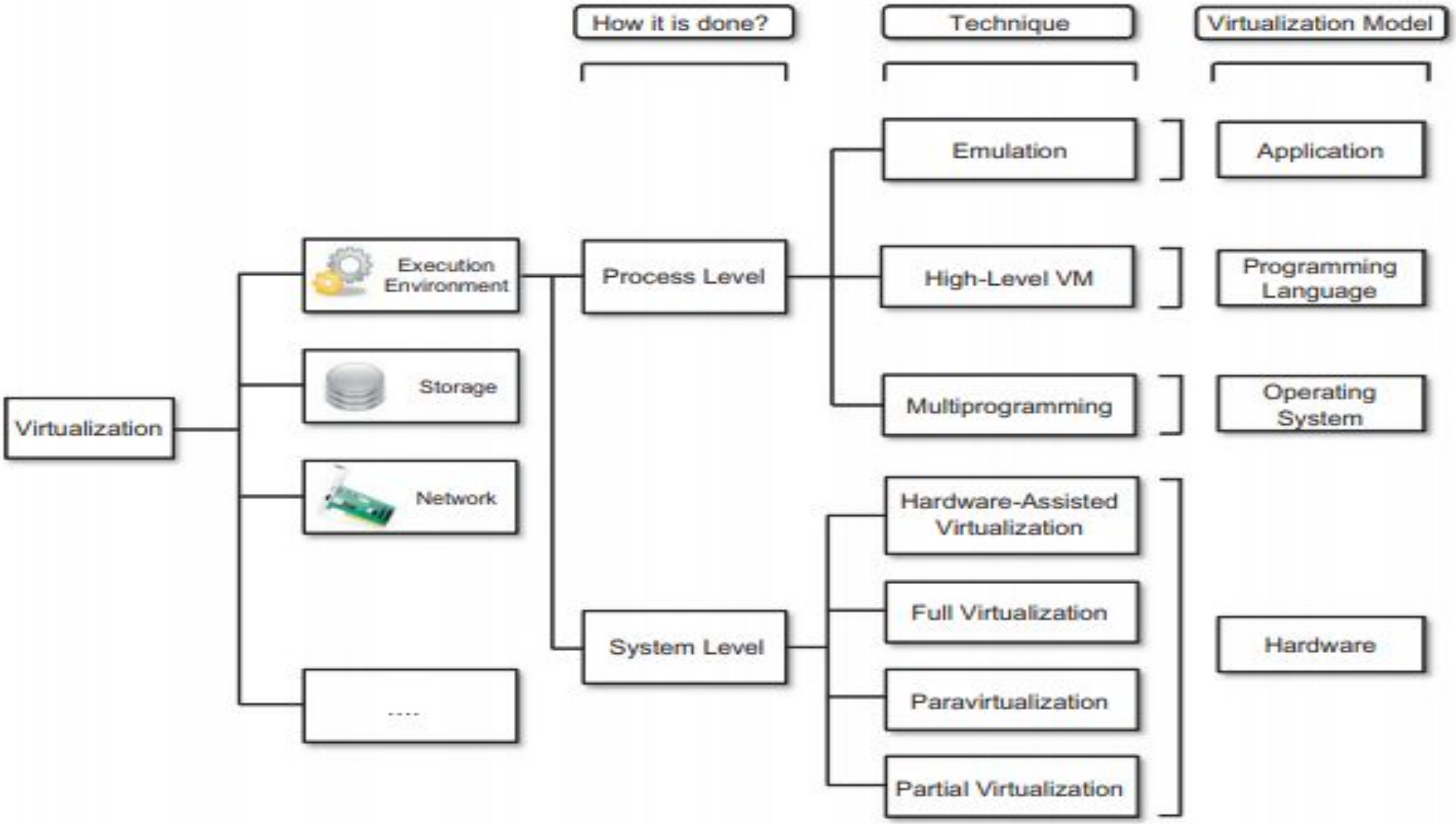
- Virtualization allows providing guests whether they are operating systems, applications, or other entities with a completely separate environment, in which they are executed.
- The guest program performs its activity by interacting with an **abstraction layer**, which provides access to the underlying resources.
- The **virtual machine can filter the activity of the guest and prevent harmful operations against the host**.

3. **Portability**

- The concept of ***portability*** **applies** in different ways according to the specific type of virtualization considered.
- In the case of a **hardware virtualization solution, the guest is packaged into a virtual image that**, in most cases, can be safely moved and **executed on top of different virtual machines**.
- Except for the file size, this happens with the same simplicity with **which we can display a picture image in different computers.**
- Virtual images are generally copyrighted formats that require a specific virtual machine manager to be executed.

Taxonomy of virtualization techniques

- Virtualization covers a wide range of **emulation techniques** that are applied to different areas of computing.
- A classification of these techniques **helps us** better understand their characteristics and use.
- Figure shows the A taxonomy of virtualization techniques


How it is done?
Technique
Virtualization Model
Virtualization
Execution Environment
Storage
Network
....
Process Level
System Level
Emulation
High-Level VM
Multiprogramming
Hardware-Assisted Virtualization
Full Virtualization
Paravirtualization
Partial Virtualization
Application
Programming Language
Operating System
Hardware

- The **first classification discriminates against the service or entity that is being emulated**.
- Virtualization is mainly used to emulate ***execution environments***, ***storage*, and *networks*.**
- Among these categories, *execution virtualization* constitutes the oldest, most popular, and most developed area
- In particular we can divide these execution virtualization techniques into **two major categories** by considering the type of host they require.
  - ***Process-level***
  - ***System-level***

- **Process-level** techniques are implemented on **top of an existing operating system**, which has full control of the hardware.
- ***System-level*** techniques are **implemented directly on hardware** and do not require—or require a minimum of support from—an existing operating system.
- Within these two categories **we can list various techniques that offer the guest a different type of virtual computation environment.**
- Like bare hardware, operating system resources, low-level programming language, and application libraries.

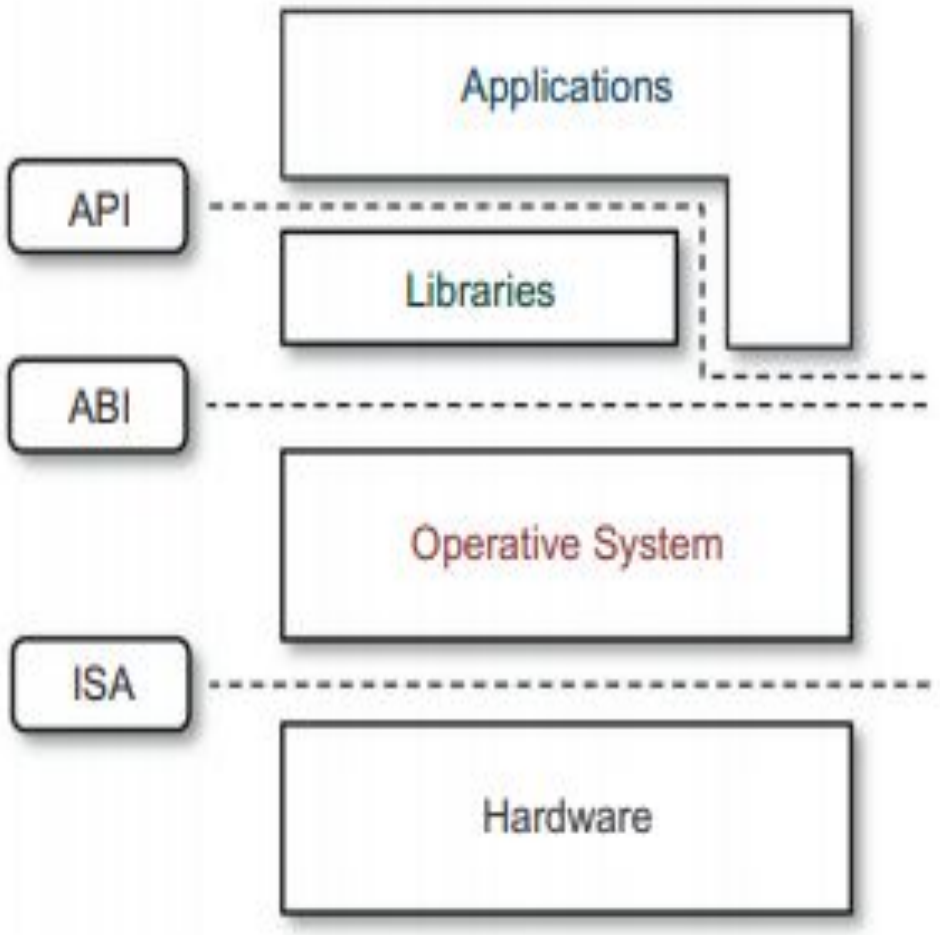
3.3.1 Execution virtualization

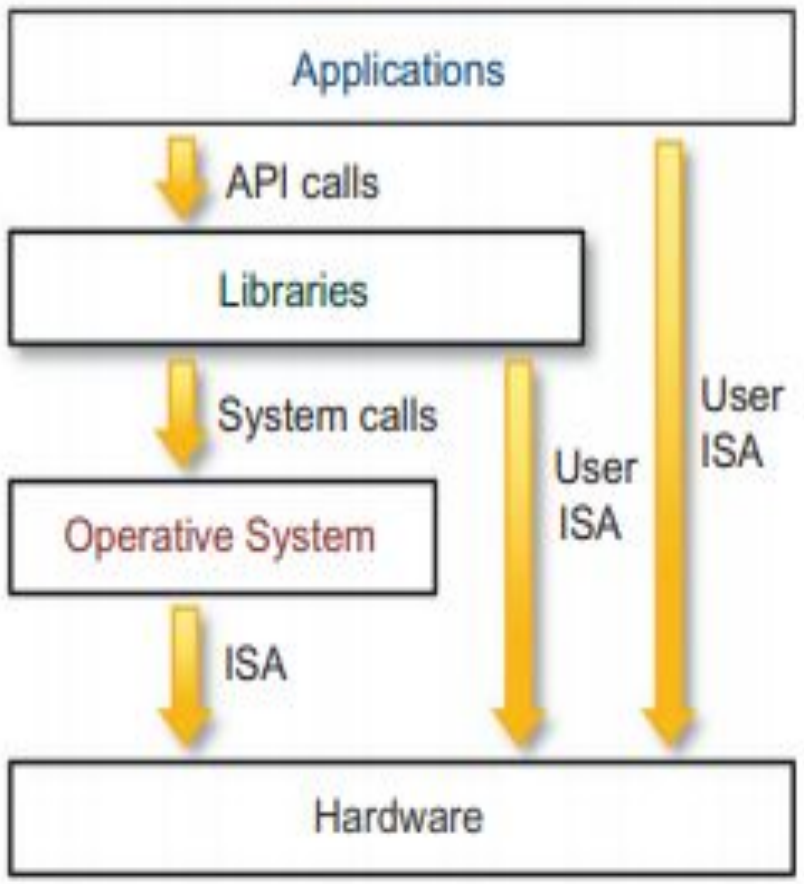
- ***Execution virtualization*** includes **all techniques** that **aim to emulate** an **execution environment** that is separate from the **one hosting the virtualization layer.**
- All these techniques **concentrate their interest** on providing support for the **execution of programs**.
- whether these are the **operating system**, a **binary specification** of a program compiled against an abstract machine model, or **an application**.
- Therefore, execution virtualization can be **implemented directly on top of the hardware** by the **operating system, an application, or libraries dynamically or statically** linked to an application image.

Machine reference model

- Virtualizing an execution environment at different levels of the **computing stack requires a reference model**.
- This reference model defines the **interfaces** between the **levels of abstractions**.
- From this perspective, **virtualization techniques actually replace one of the layers and intercept the calls that are directed toward it.**
- Modern computing systems can be expressed in terms of the reference model described in Figure(next slide).

A Machine Reference Model





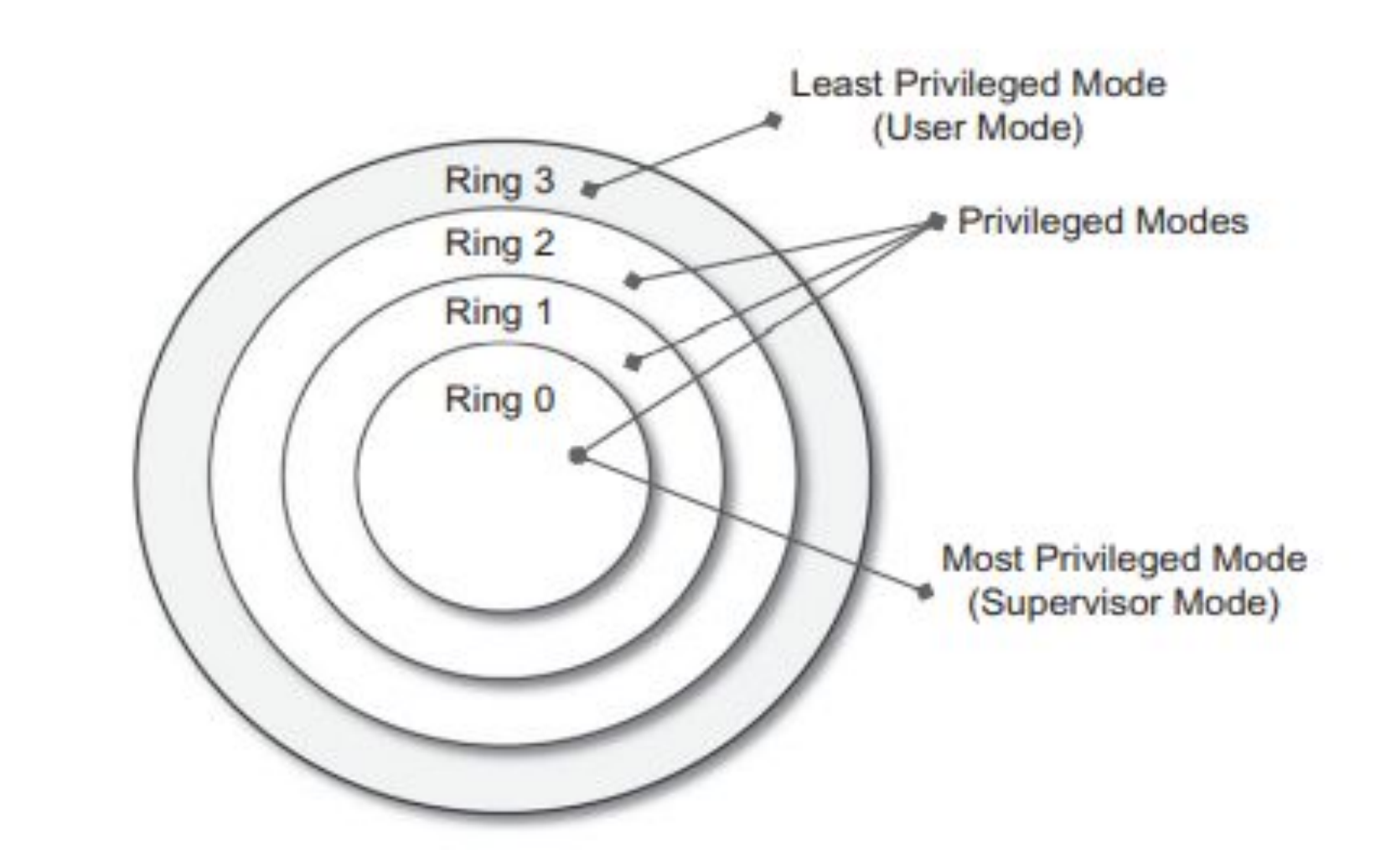
- At the **bottom layer**, the model for the hardware is expressed in terms of the ***Instruction Set Architecture (ISA)*.**
- Which defines **the instruction set for the processor, registers, memory, and interrupts management.**
- **ISA is the interface between hardware and software**.
- It is important to the **operating system** (OS) developer (*System ISA*) and **developers of applications** that directly manage the underlying hardware (*User ISA*).
- This interface(ISA) allows portability of applications and libraries across operating systems that implement the same application binary interfaceABI.

- The highest level of abstraction is represented by the application programming interface(**API**).
- Which interfaces **applications to libraries and/or the underlying operating system.**
- The application binary interface(ABI) separates the **operating system layer from the applications and libraries, which are managed by the OS.**
- ABI covers details such as **low-level data types, alignment, and call conventions and defines a format for executable programs.**
- System calls are defined at this level.

- The **instruction set** exposed by the hardware has been divided into **different security classes** that define who can operate with them.
- The **first distinction** can be made **between privileged and Nonprivileged instructions.**
- **Nonprivileged instructions** are those instructions that can be used **without interfering** with other tasks because they **do not access shared resources.**
- for example, all the **floating, fixed-point**, and **arithmetic instructions.**
- **Privileged instructions** are those that are **executed under specific restrictions.**
- Are mostly used for sensitive operations, which expose (behavior-sensitive) or modify (control-sensitive) the privileged state.

- For instance, **behavior-sensitive instructions are those that operate on the I/O**,
- whereas **control-sensitive instructions alter the state of the CPU registers.**
- A possible implementation features a hierarchy of privileges (Figure in next slide) in the form of **Ring-based Security**.
- ***Ring 0*, *Ring 1*, *Ring 2*, and *Ring 3***;
- Ring 0 is in the most **privileged level** and Ring 3 in the **least privileged level.**
- **Ring 0 is used by the kernel of the OS**,
- **Rings 1 and 2 are used by the OS-level services, and**
- **Ring 3 is used by the user**.
- Recent systems support only two levels, with Ring 0 for supervisor mode and Ring 3 for user mode.

Security Rings and Privileged mode

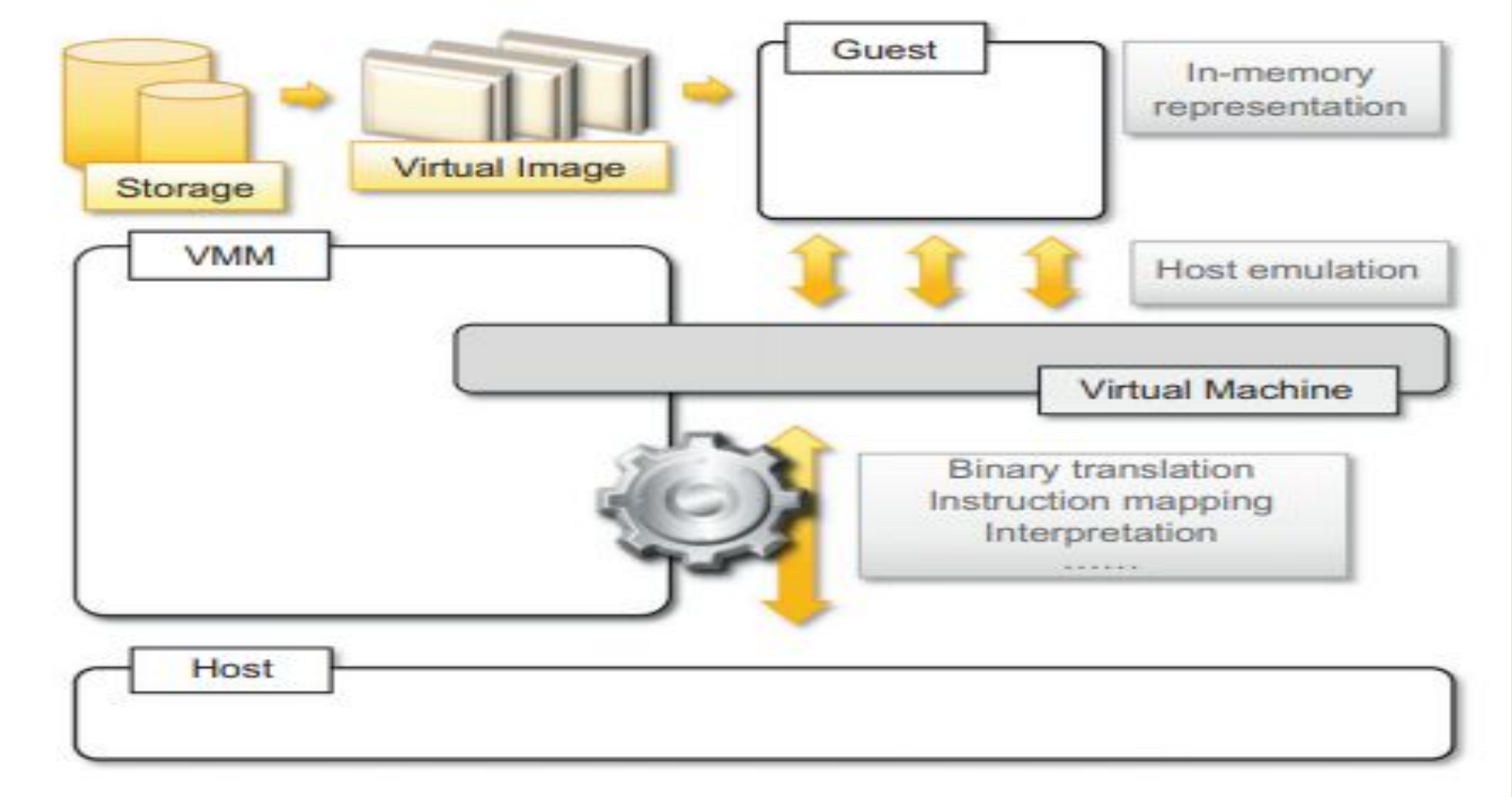


- All the current systems support at least two different execution modes: ***supervisor mode*** **and** ***user mode***.
- The **first mode** denotes an **execution mode** in which all the instructions (privileged and Nonprivileged) can be executed without any restriction.
- In **execution mode**, also called ***master mode*** **or** ***kernel mode***, is generally used by the operating system (or the hypervisor) to perform **sensitive operations on hardware level resources**.
- In **user mode**, there are restrictions to control the machine-level resources.
- Conceptually, the hypervisor runs above the **supervisor mode.**

Hardware-level virtualization

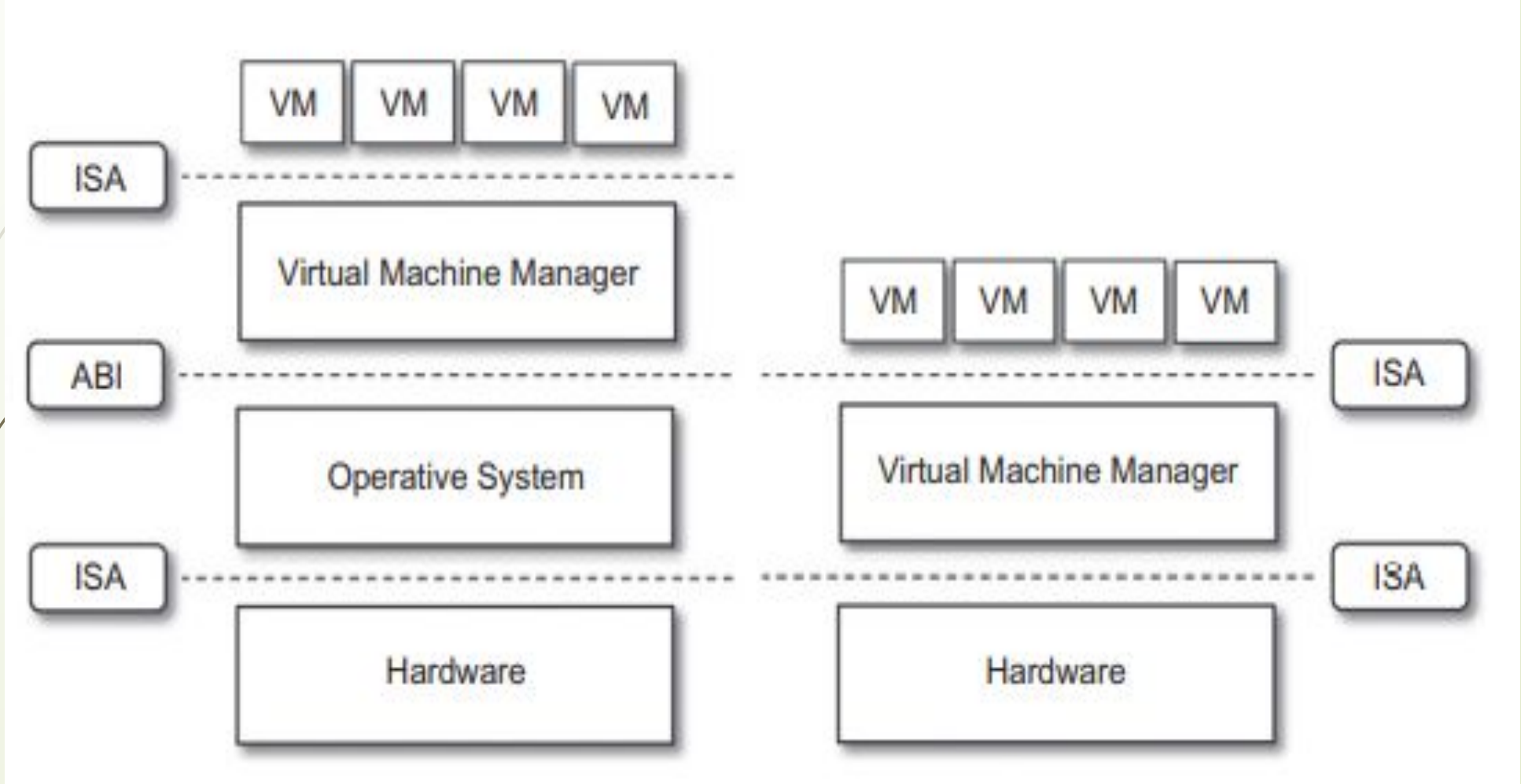
- Virtualization technique that provides an **abstract execution environment in terms of computer hardware**. on top of which a **guest** operating system can be run.
- In this model, the **GUEST is represented by the operating system**,
- The **HOST by the physical computer hardware**,
- The **VIRTUAL MACHINE by its emulation**, and
- The **VIRTUAL MACHINE MANAGER by the hypervisor** (see Figure in next slide).
- The hypervisor is generally **a program or a combination of software and hardware** that allows the abstraction of the underlying physical hardware.
- **Hardware-level virtualization is also called *system virtualization***, since it provides **ISA** to virtual machines, which is the representation of the hardware interface of a system.

Figure: Hardware virtualization reference model.



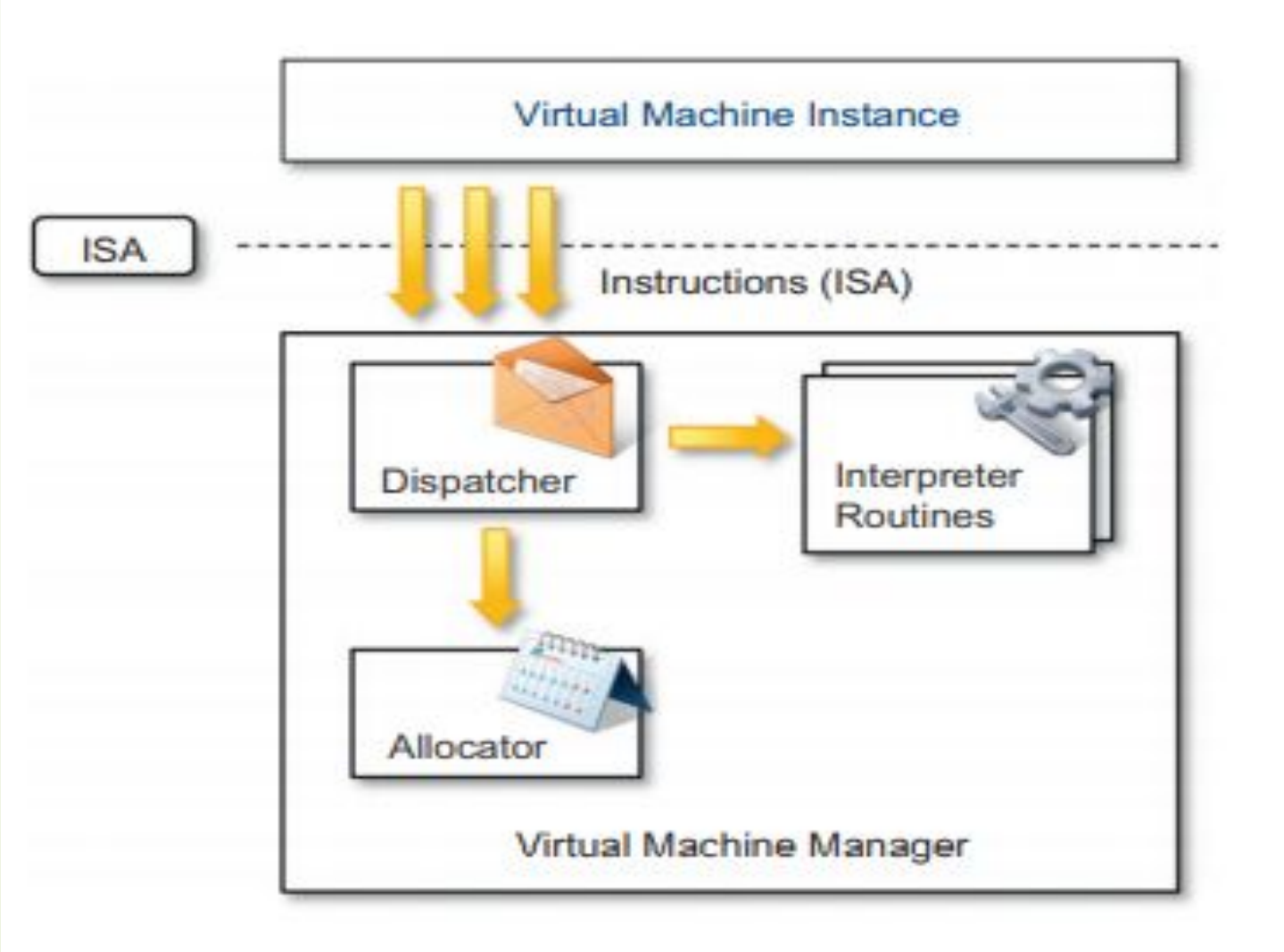
- **Hypervisors**: A fundamental element of hardware virtualization is the **hypervisor, or virtual machine manager (VMM).**
- It **recreates** a hardware environment in which **guest operating systems are installed.**
- There are two major types of hypervisor: ***Type I*** and ***Type II*** (see Figure in next slide).
- ***Type I*** hypervisors run directly on **top of the hardware**.
- Therefore, they take the place of the operating systems and **interact directly with the ISA interface** exposed by the underlying hardware
- They **emulate this interface in order to allow the management of guest operating systems.**
- This type of hypervisor is also called a ***native virtual machine*** since it runs natively on hardware.

A graphical representation of two types of Hypervisors.



- **Type II hypervisors** require the support of an operating system to provide virtualization services.
- This means that they are **programs managed by the operating system**
- Which interact with it through the **ABI** and emulate the **ISA** of virtual hardware for guest operating systems.
- This type of hypervisor is also **called a hosted virtual machine** since it is hosted within an operating system.
- A **virtual machine manager** is internally organized as described in Figure(next slide).
- Three main modules, **Dispatcher, Allocator, and Interpreter**, coordinate their activity in order to emulate the underlying hardware.

A hypervisor reference architecture



- The **dispatcher constitutes the entry point of the monitor** and **reroutes the instructions** issued by the **virtual machine** instance to one of the two other modules.
- The **allocator** is responsible for deciding the system resources to be provided to the VM:
- whenever a virtual machine tries to execute an instruction that results in changing the machine resources associated with that VM,
- The **allocator is invoked by the dispatcher**.
- The **interpreter** module consists of interpreter routines.
- These are executed whenever a virtual machine executes a privileged instruction:
- A trap is triggered and the corresponding routine is executed.

- **Three properties** of Virtual Machine Manager that have to be satisfied:

• **Equivalence.** A guest running under the control of a **virtual machine manager** should exhibit the same behavior as when it is executed directly **on the physical host**.

• **Resource control.** The virtual machine manager should be in complete control of **virtualized resources**.

• **Efficiency.** A statistically dominant fraction of the machine instructions should be executed without intervention from the virtual machine manager.

Virtualization techniques

1. **Hardware level virtualization techniques**
   - Hardware-assisted virtualization
   - Full virtualization.
   - Paravirtualization.
   - Partial virtualization.
2. **Operating system-level virtualization**
3. **Programming language-level virtualization**
4. ***Application-level virtualization***
5. **Other types of virtualization**
   - Storage virtualization
   - Network virtualization
   - Desktop virtualization
   - Application server virtualization

1. Hardware level virtualization techniques

A. Hardware-assisted virtualization:

- This term refers to a scenario in which the hardware provides **architectural support** for building a **VMM** able to run a **guest operating system** in complete isolation.
- This technique was originally introduced in the **IBM System/370**.
- At present, examples of hardware-assisted virtualization are the extensions to the x86-64 bit architecture introduced with *Intel VT*.
- Products such as VMware Virtual Platform, introduced in 1999 by **VMware**, which pioneered the field of x86 virtualization, were based on this technique.
- After 2006, **Intel and AMD** introduced processor extensions, and a wide range of virtualization solutions took advantage of them:
- Kernel-based Virtual Machine (KVM), VirtualBox, Xen, VMware, Hyper-V, Sun xVM, Parallels, and others.

B. Full virtualization.

- ***Full virtualization*** **refers to the ability to run a program, most likely an operating system, directly on top of a virtual machine and without any modification.**
- To make this possible, virtual machine managers are required to provide a **complete emulation of the entire underlying hardware.**
- The principal **advantage** of full virtualization is **complete isolation**, which leads to **enhanced security**, **ease of emulation** of **different architectures**, and **coexistence of** different systems on the same platform.
- A simple solution to achieve full virtualization is to provide a **virtual environment for all the instructions**, thus posing some limits on performance.

C. Paravirtualization.

- This is a not-transparent virtualization solution that allows implementing thin **Virtual Machine Managers**.
- **Paravirtualization techniques expose a software interface to the virtual machine that is slightly modified from the host and, as a consequence, guests(OS) need to be modified.**
- The aim of Paravirtualization is **to provide** the **capability** to **demand the execution of performance-critical operations directly on the host.**
- This technique has been successfully used by Xen for providing virtualization solutions for Linux-based operating systems specifically ported to run on Xen hypervisors.

D. Partial virtualization

- Partial virtualization provides **a partial emulation** of the **underlying hardware**, thus not allowing the complete **execution of the guest operating system in complete isolation**.
- **Partial virtualization allows many applications to run transparently, but not all the features of the operating system can be supported,** as happens with full virtualization.
- Partial virtualization was implemented on the experimental **IBM M44/44X**. Address space virtualization is a common feature of contemporary operating systems.

2. Operating system-level virtualization

- Operating system-level virtualization offers the opportunity to **create different and separated execution environments for applications** that are managed concurrently.
- Differently from hardware virtualization, there is **no virtual machine manager or hypervisor.**
- The virtualization is **done within a single operating system, where the OS kernel allows for multiple isolated user space instances**.
- The **kernel is also responsible for sharing the system resources** among instances and for limiting the impact of instances on each other.

- A user space instance in general contains a proper view of **the file system**, which is completely isolated, and **separate IP addresses**, **software configurations**, and **access to devices**.
- Operating system-level virtualization **aims to provide separated and multiple execution containers** for running applications.
- Compared to hardware virtualization, this strategy imposes **little or no overhead** because applications directly use **OS system calls** and there is **no need for emulation.**

3. Programming language-level virtualization

- Programming language-level virtualization is mostly used to achieve ease of **deployment of applications**, **managed execution**, and **portability across** different **platforms and operating systems**.
- It consists of a **virtual machine** executing the **byte code of a program(also termed portable code or p-code),** which is the result of the **compilation process.**
- **Bytecode is object-oriented programming (OOP) code compiled to run on a virtual machine (VM)**
- **Compilers implemented and used this technology to produce a binary format** representing the machine code for an abstract architecture.

- The characteristics of this architecture vary from implementation to implementation
- Programming language-level virtualization has a long trail in computer science history.
- **Originally was used in 1966** for the implementation of **Basic Combined Programming Language (BCPL),** a language for writing compilers and one of the **ancestors of the C programming language.**
- Other important examples of the use of this technology have been the **UCSD Pascal and Smalltalk**.
- Virtual machine programming languages become popular again with **Sun‟s introduction of the Java platform in 1996.**

- Currently, the **Java platform and .NET Framework** represent the most popular technologies for enterprise application development.
- The main advantage of programming level virtual machines, also called ***process virtual machines*****,** is the ability to provide a **uniform execution environment across different platforms**.
- **Programs compiled into byte code can be executed on any operating system and platform** for which a virtual machine able to execute that code has been provided.

Application-level virtualization

- Application-level virtualization is a technique allowing **applications to be run in runtime environments** that do not natively support all the features required by such applications.
- In general, these techniques are mostly concerned with **partial file systems, libraries, and operating system component emulation.**
- Such emulation is performed by a **thin layer—a program or an operating system component**—that is **in charge of executing the application.**
- Emulation can also be used to execute program binaries compiled for different hardware architectures.

In this case, one of the following strategies can be implemented:

- **Interpretation.** In this technique every source instruction is interpreted **by an emulator** for executing native ISA instructions, leading **to poor performance**.

  Interpretation has a minimal startup cost but a huge overhead, since each instruction is emulated.

- **Binary translation.** In this technique every source instruction **is converted to native instructions with equivalent functions**. After a block of instructions is translated, it is cached and reused.

  Binary translation has a large initial overhead cost, but over time it is subject to better performance, since previously translated instruction blocks are directly executed.

- Application virtualization is a good solution in the case of **missing libraries in the host operating system**.
- One of the most popular solutions implementing **application virtualization is Wine.**
- **Wine** is a **free and open-source compatibility layer** that aims to allow **application software and computer games** developed for Microsoft Windows to run on Unix-like operating systems.
- No code emulation or virtualization occurs when running a Windows application under Wine**.**
- Which is a software application allowing **Unix-like operating systems to execute programs written for the Microsoft Windows platform**

Virtualization techniques

1. **Hardware level virtualization techniques**
2. **Operating system-level virtualization**
3. **Programming language-level virtualization**
4. ***Application-level virtualization***
5. **Other types of virtualization**
   - Storage virtualization
   - Network virtualization
   - Desktop virtualization
   - Application server virtualization

Other types of virtualization

1. Storage virtualization

- Storage virtualization is a **system administration** practice that allows **decoupling the physical organization of the hardware from its logical representation.**
- Using this technique, **users do not have to be worried about the specific location of their data**, which can be identified using a logical path.
- There are different techniques for storage virtualization,
- The most popular being network-based virtualization are

  Storage area networks (SANs) &

  Network-attached storage (NAS)

2. Network virtualization

- *Network virtualization* combines **hardware appliances and specific software for the creation and management of a Virtual network.**
- Network virtualization can **aggregate** different physical networks into

1.Single logical network (*external* network virtualization)

2.Network-like functionality to an operating system partition (*internal* network virtualization).

- The result of external network virtualization is generally a ***virtual LAN (VLAN).***
- A VLAN is an aggregation of hosts that communicate with each other as though they were located under the same broadcasting domain.

- There are several options for implementing internal network virtualization
- The **guest can share the same net- work interface** of the host and **use Network Address Translation (NAT)** to access the network;

3. Desktop virtualization

- Desktop virtualization **abstracts the desktop environment available on a personal computer** in order to provide access to it using a client/server approach.
- Desktop virtualization provides the **same outcome** of hardware virtualization but serves a different purpose.
- Desktop virtualization **makes accessible a different system as though it were natively installed on the host**, but this system is remotely stored on a different host and accessed through a network connection.

- Moreover, desktop virtualization **addresses the problem of making the same desktop environment accessible from everywhere.**
- The advantages of desktop virtualization are
  - **High availability,**
  - **Persistence,**
  - **Accessibility, and**
  - **Ease of management.**
- Infrastructures for desktop virtualization based on cloud computing solutions include **Sun Virtual Desktop Infrastructure (VDI), Parallels Virtual Desktop Infrastructure (VDI), Citrix XenDesktop, and others**.

4. **Application server virtualization**

- Application server virtualization abstracts of collection of **application servers that provide the same services as a single virtual application server** by using load-balancing strategies.
- By providing a **high-availability infrastructure** for the services hosted in the application server.
- This is a particular form of virtualization and serves the same purpose of storage virtualization.
- Providing a better quality of service rather than emulating a different environment.
- **Ex: FreeVPS, Linux Vserver, and OpenVZ**

Virtualization and cloud computing

- Virtualization plays an important role in **cloud computing** since it allows for the **appropriate degree of customization**, **security**, **isolation**, and **manageability** that are fundamental for delivering **IT services on demand.**
- Virtualization technologies are primarily used to offer **configurable computing environments and storage**.
- Particularly important is the role of **virtual computing environment and execution virtualization techniques**
- Among these, **hardware** and **programming language virtualization** are the techniques adopted in **cloud computing** systems
- Hardware virtualization is an **enabling factor for solutions** in the **Infrastructure-as-a-Service (IaaS)** market segment,

- **Programming language virtualization** is a technology leveraged in **Platform-as-a-Service (PaaS)** offerings
- **customizable** and sandboxed environment constituted an **attractive business** opportunity for companies featuring a **large computing infrastructure** that was able to **sustain** and **process huge workloads**.
- virtualization also **allows isolation and a finer control**, thus simplifying the **leasing of services** and **their accountability** on the vendor side.
- Virtualization allows us to **create isolated and controllable environments**, it is possible to serve these environments with the same resource without them interfering with each other.

- If the underlying **resources are capable enough**, there will be **no evidence of such sharing**
- It allows **reducing the number** of active resources **by aggregating virtual machines** over a smaller number of resources that become fully utilized.
- This practice is also known as ***server consolidation***,
- while the movement of virtual machine instances is called ***virtual machine migration***.

Pros and cons of virtualization

Advantages of virtualization

1. **Managed execution and isolation:**

- Allow building **secure and controllable** computing environments.
- **Allocation of resources** and their partitioning among different guests is simplified, being the virtual host controlled by a program.
- This enables **fine-tuning of resources**, which is very important in a server consolidation scenario and is also a requirement for effective quality of service

2. Portability

- **Portability and self-containment** also contribute to reducing the costs of maintenance.
- Portability and self-containment **simplify their administration**. Java programs are ***"compiled once and run every where"*** ; they **only require that the Java virtual machine** be installed on the host.
- By means of **virtualization it is possible to achieve** a more **efficient use of resources**.
- Multiple systems can **securely coexist and share** the resources of the underlying host, **without interfering with each other.**

- This is a prerequisite for **server consolidation**, which allows **adjusting the number of active physical resources dynamically**.
- According to the current load of the system, thus creating the opportunity **to save in terms of energy consumption and to be less impacting on the environment.**
- Finally, by means of **virtualization it is possible to achieve a more efficient use of resources.**

Disadvantages

1. Performance degradation

- The most evident is represented by a **performance decrease** of guest systems as a result of the intermediation performed by the virtualization layer.
- Inefficient utilization of the **host or a degraded** user experience
- Performance is definitely one of the **major concerns** in using virtualization technology. Since **virtualization interposes an abstraction layer between the guest and the host, the guest can experience increased latencies.**

- The **overhead** introduced in hardware virtualization by the following activities
  - Maintaining the status of virtual processors
  - Support of privileged instructions (trap and simulate privileged instructions)
  - Support of paging within VM
  - Console functions
- In the case of programming language virtual machines (Java, .NET, and others). **Binary translation and interpretation can slow down the execution of managed applications**.
- Moreover, because their execution is filtered by the runtime environment, **access to memory and other physical resources can represent sources of performance degradation.**

2. Inefficiency and degraded user experience

- Virtualization can sometime lead to an **inefficient use of the host.**
- In particular, some of the **specific features** of the host **cannot be exposed** by the abstraction layer and then become **inaccessible.**
- In the case of **hardware virtualization**, this could happen for device drivers:
- The virtual machine can sometime simply provide a default graphic card that maps only a subset of the features available in the host.
- In the case of **programming-level virtual machines**, some of the features of the underlying operating systems may become **inaccessible unless specific libraries are used.**

3. Security holes and new threats

- Virtualization opens the **door to a new and unexpected form of *phishing*.**
- The capability of **emulating a host** in a completely transparent manner led the way to malicious programs that are **designed to extract sensitive information from the guest.**
- In the case of hardware virtualization, **malicious programs can preload themselves before the operating system** and **act as a thin virtual machine manager toward it.**
- The **operating system** is then **controlled** and can be **manipulated to extract sensitive information of interest to third parties**

Review questions

1. What is virtualization and what are its benefits?
2. What are the characteristics of virtualized environments?
3. Discuss classification or taxonomy of virtualization at different levels.
4. Discuss the machine reference model of execution virtualization.
5. What are hardware virtualization techniques?
6. List and discuss different types of virtualization.
7. What are the benefits of virtualization in the context of cloud computing?
8. What are the disadvantages of virtualization?
9. What is Xen? Discuss its elements for virtualization.
10. Discuss the reference model of full virtualization.
11. Discuss the architecture of Hyper-V. Discuss its use in cloud computing.